Heavy-Duty Hydrogen Station Equipment Performance Device (HD HyStEP) Specifications and Design Considerations

Spencer Gilleon, Jacob Thorson, Taichi Kuroki, and Sam Sprik

National Renewable Energy Laboratory

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1 Introduction

The original Hydrogen Station Equipment Performance (HyStEP) device was commissioned in 2015 and was critical for the rapid validation of light-duty (LD) hydrogen fueling stations. As applications for hydrogen as a heavy-duty (HD) transportation fuel continue to grow, new heavy-duty stations are being developed to fuel these vehicles that require larger onboard storage tank systems to meet HD transportation demands and drive cycles. The differences in size and geometry from LD vehicles have driven the creation of new hydrogen fueling protocols that will enable safe and economical fueling of HD vehicles.

With the new requirements that are set out in HD fueling protocols like SAE J601-5, a new HyStEP-like device is needed to evaluate the capabilities of high-flow hydrogen stations to fuel HD vehicles to the new protocol standard. To create a heavy-duty HyStEP (HD HyStEP) device, an effort has been undertaken to evaluate the requirements that would form the basis for the design of a successful HD HyStEP device. The primary design goal of this HD HyStEP device is its capability of following a test methodology similar to what is outlined in CSA HGV 4.3, which guides the validation of LD fueling dispensers but with adjustments to verify adherence to the HD fueling protocols (SAE J2601-5) for 70 MPa and 35 MPa pressure class vehicles rather than the LD fueling protocols (SAE J2601). A dedicated test method for HD fueling dispensers, CSA HGV 4.3.5, is undergoing development and is anticipated to be published in 2026. The design presented here aims to provide information that enables the creation of an HD HyStEP device that achieves the primary design goal while being informed by the years of experience from the current HyStEP operators.

2 Design Specifications

A thorough review of the SAE J2601-5 technical information report (TIR) was conducted to identify the impacts that the fueling protocols would have on the validation process for an HD hydrogen fueling station. A primary difference identified between the existing LD protocols and the new HD protocols was the increased granularity in the compressed hydrogen storage system (CHSS) capacity and the additional considerations given to the size of the largest individual hydrogen storage tank (TVL). These changes require a more complex hydrogen storage system onboard the HD HyStEP device, so that a representative variety of fueling conditions may be validated by reconfiguring the internal tank connections. Device piping and associated connection hardware must be sized appropriately to ensure the device's capability of handling all flow rates required by HD fueling protocols. Representative piping diameters and connector sizes are presented in Appendix A and are designed to accommodate flow rates as high as 300 g/s and as low as 60 g/s. Some of the key design specifications that were identified during this review process are shown in the following subsections. For more detailed information, please refer to the primary documents included with this report. Please also note that this design effort has been focused on the vehicle-side hydrogen system, and additional engineering effort will be required for the safe design of the trailer and other non-hydrogen support equipment during the detailed design effort.

2.1 General System

- The system will be mobile and designed to be transported to high-flow HD fueling stations of varied design and manufacture.
- The system will include the valves and piping needed to allow for simultaneous fueling and defueling of separate storage vessels.
- The system must support different flow rates, per SAE J2601-5.
- The system will include an external human-machine interface (HMI) for active control and monitoring during testing. In addition to the HMI, external indicators will be included to visually indicate the status of the HD HyStEP device (e.g., normal, gas detection, alarm, etc.).
- The system will include a separate, stand-alone vent mast with hose connection that will allow for process venting of hydrogen independent of the station's vent system. As this testing is likely to consume hundreds of kilograms of hydrogen, the system has been designed to enable the safe recirculation of hydrogen from the HD HyStEP device to the hydrogen station via a typical gaseous hydrogen filling stanchion. Note that although this may be conducted safely, factors regarding hydrogen purity and other logistical questions must be addressed prior to this recirculation activity.
- Independent of the stand-alone vent mast, the system will include permanently installed safety vents that will be routed to the exterior rooftop of the HD HyStEP device trailer. These vents will include weather caps to prevent the ingress of water or debris without preventing the safe release of gas in the case of a safety device opening (e.g., pressure safety valve [PSV], thermally activated pressure relief device [TPRD], etc.).
- The system will be enclosed for weather protection and will include active, monitored ventilation in all enclosed spaces that contain compressed hydrogen.
- The ventilation will include interlocks to shut down in the case of a loss of ventilation to reduce the likelihood of hydrogen accumulation in the case of an undetected leak.
- The interior of the trailer will be constructed of and/or clad in non-flammable materials to prevent the spread of fire.
- The trailer doors will be lockable from the exterior to prevent unauthorized access. These doors will be able to be opened from the interior when locked or unlocked.

2.2 Hydrogen Storage Vessels

- All onboard hydrogen storage vessels will be Type IV for the temperature and pressure ratings of 85°C and 87.5 MPa, complying with the requirements for onboard storage system temperature and pressure prescribed in the SAE J2601-5 fueling H70HF protocol.
- The CHSS will be composed of user-selectable storage volumes enabling configurations ranging in total volume from 250–3,000+ L.
- The individual hydrogen storage tanks will range in size from 50–350+ L and will include multiple tanks with volumes each in the ranges of 50–200 L, 200–350 L, and >350 L.
- The individual hydrogen tank volumes will be selected to maximize the number of unique combinations of total CHSS volume and largest tank volume (TVL) category as defined in SAE J601-5 (see below in Table 1).

Table 1. CHSS and TVL category combinations table with the fueling categories that can be configured using the example design outlined in this document. A similar table should be developed for the final, detailed HD HyStEP device design.

	CHSS Category (L)							
TVL (L)	248.6	500	1,000	1,500	2,000	2,500	3,000	5,000
50 ≤ TVL ≤ 200	X	X						
200 < TVL ≤ 350	X	X	X	X				
350 < TVL ≤ 1,000	N/A	X	X	X	X	X	X	

2.3 Hydrogen Fueling Receptacles

- The system will include an H70 LD fueling receptacle meeting the requirements of SAE J2600.
- The system will also include an H70 high-flow receptacle per relevant standards when they are published. If this does not exist at the time of fabrication, supporting equipment will be included to simplify the installation of an H70 high-flow receptacle when it becomes publicly available.
- The system will include an H35 HD (high flow) receptacle per relevant standards.
- The system will include IrDA wireless communication hardware for each receptacle to allow for communication between the HD HyStEP device and fueling dispenser per the applicable standards (e.g., SAE J2799).

2.4 Dispensed Mass Metering

Currently, there are no dispensed mass metering devices available for HD fueling. A mass flow meter designed for HD high-flow fueling is an option for accurately measuring the dispensed mass. These meters can measure the instantaneous mass flow at a time resolution of approximately 1.0 millisecond with an accuracy of 0.5%, meaning that the integrated value of the measured mass flow rates would be nearly identical to the mass dispensed by the station to the HD HyStEP device. However, calibration and verification are also needed for these devices to ensure these resolutions and accuracies are within specification for use at various stations.

2.5 Particulate Matter and Contaminant Measurement

For HD hydrogen stations to support fuel cell vehicle refueling, assurance of high purity dispensed hydrogen is required. Fuel cell materials are highly sensitive to chemical impurities that can originate from the hydrogen station or from delivered hydrogen sources. While the HD HyStEP device does not have a fuel cell onboard and does not necessarily require a specified hydrogen quality, it may be beneficial for the device to have onboard hydrogen contaminant detection (HCD) or obtain particulate matter sampling to monitor hydrogen purity both being dispensed into the HD HyStEP device and any hydrogen recirculated back to the station from the

HD HyStEP device. This section briefly describes the hydrogen quality requirements and high-level considerations for HCD.

Standards such as Compressed Gas Association (CGA) G-5.3 Commodity Specification for Hydrogen and SAE J2719 – Hydrogen Fuel Quality for Fuel Cell Vehicles specify contaminant type and maximum concentration requirements for both gaseous and liquid hydrogen. Current station fuel quality verification processes are based on sporadic collection of gas samples using ASTM methods (e.g., D7650 and D7606) and remote analysis at a testing laboratory, which can be time consuming when immediate analysis for hydrogen quality is needed.

Implementation of an HCD device tied into HD HyStEP local vent lines and/or recirculation lines upstream of the station could prove beneficial for understanding the hydrogen quality both dispensed to the HD HyStEP and protect the station from any potential contaminants in the hydrogen being recirculated back to the station originating from the HD HyStEP device. An HCD device located on local vent lines or within recirculation piping can provide near-real-time verification of hydrogen quality and without disrupting the fueling validation process. Any detection devices located upstream of the dispenser or in-line with piping upstream of the HD HyStEP CHSS may need additional pressure drop calculations to understand any potential effects it may cause during the fueling process.

2.6 Maximum Hydrogen Venting and Tank Depressurization Rates

This section details the maximum venting and depressurization rates required when depressurizing the HD HyStEP CHSS either via hydrogen recirculation to the station or local venting to atmosphere. These rates help prevent the storage tank inner liner materials from experiencing temperatures below the lower limit of -40°C. Temperatures any lower can cause damage to the CHSS tank materials and other sensors installed on the HD HyStEP device.

Table 2 shows the maximum hydrogen depressurization rates (MPa/min) for any of the HD HyStEP storage tanks. These rates are a function of the HD HyStEP device system temperature immediately before depressurization occurs and tank configurations. If the configuration includes a 457 L tank, the appropriate depressurization rate can be derived from the bottom row of Table 2. If the configuration does not include a 457 L tank, the rate can be derived from the top row of Table 2. As long as the hydrogen in the tank system is released at rates slower than or equal to the rates in Table 2, the inner tank temperatures should remain above the -40°C lower limit. Please refer to Appendix A for information on modeling and derivations to determine the depressurization rates. End users and HD HyStEP designers should also adhere to any depressurization rates prescribed by the tank manufacturers to ensure proper care for specific tanks. Any manufacturer prescribed rates that are slower than those given in Table 2 should supersede any recommendations given by Table 2.

Table 2. Recommended maximum depressurization rates to ensure internal tank and CHSS temperatures remain greater than -40°C

		Initial HD HyStEP System Temperature							
	TVL (L)	-20°C	-10°C	0°C	10°C	20°C	30°C	40°C	50°C
CHSS	**248.6**	0.066	0.108	0.144	0.231	0.308	0.492	0.656	0.875
	500–3,000	0.066	0.102	0.14	0.1218	0.290	0.387	0.516	0.688

3 Primary Documents

These documents are provided with the intention of acting as a starting point for a detailed design effort. Sound engineering judgement should be used when conducting that design effort. Deviations from this design should be documented (e.g., red-lined), so that they may be evaluated for their potential impact on safety and functionality.

3.1 Piping and Instrumentation Diagrams

The piping and instrumentation diagram (P&ID) in Appendix A represents the design developed as part of this effort and represents the configuration that was evaluated in the safety review that is included in Section 2.5. The design represented in this P&ID was developed to achieve a balance of maximized flexibility while remaining within practical cost and size limitations.

3.2 Bill of Materials

The bill of materials (BOM) in Appendix B is included as a starting point for a designer to understand the component quantity and specifications for the primary components that are shown in the P&ID (see Section 2.1). The design temperatures and pressures included here represent those that were assumed in the process hazard analysis (PHA) that is reported in Section 2.5. Deviations from these should be evaluated for their safety and should consider all potential operability impacts on the HD HyStEP device.

3.3 Cause and Effects Matrix

The cause and effects (C&E) matrix in Appendix C outlines the actions that are expected to occur based on a variety of operating conditions that should trigger various degrees of warnings and/or alarms. This may form the basis for a more detailed controls design consisting of a mix of hardware and software configurations. Note that some of the alarm values in this document have been left unspecified as their exact value should be determined based on the specifications of the final, detailed design.

3.4 Outline of the Operation Manual

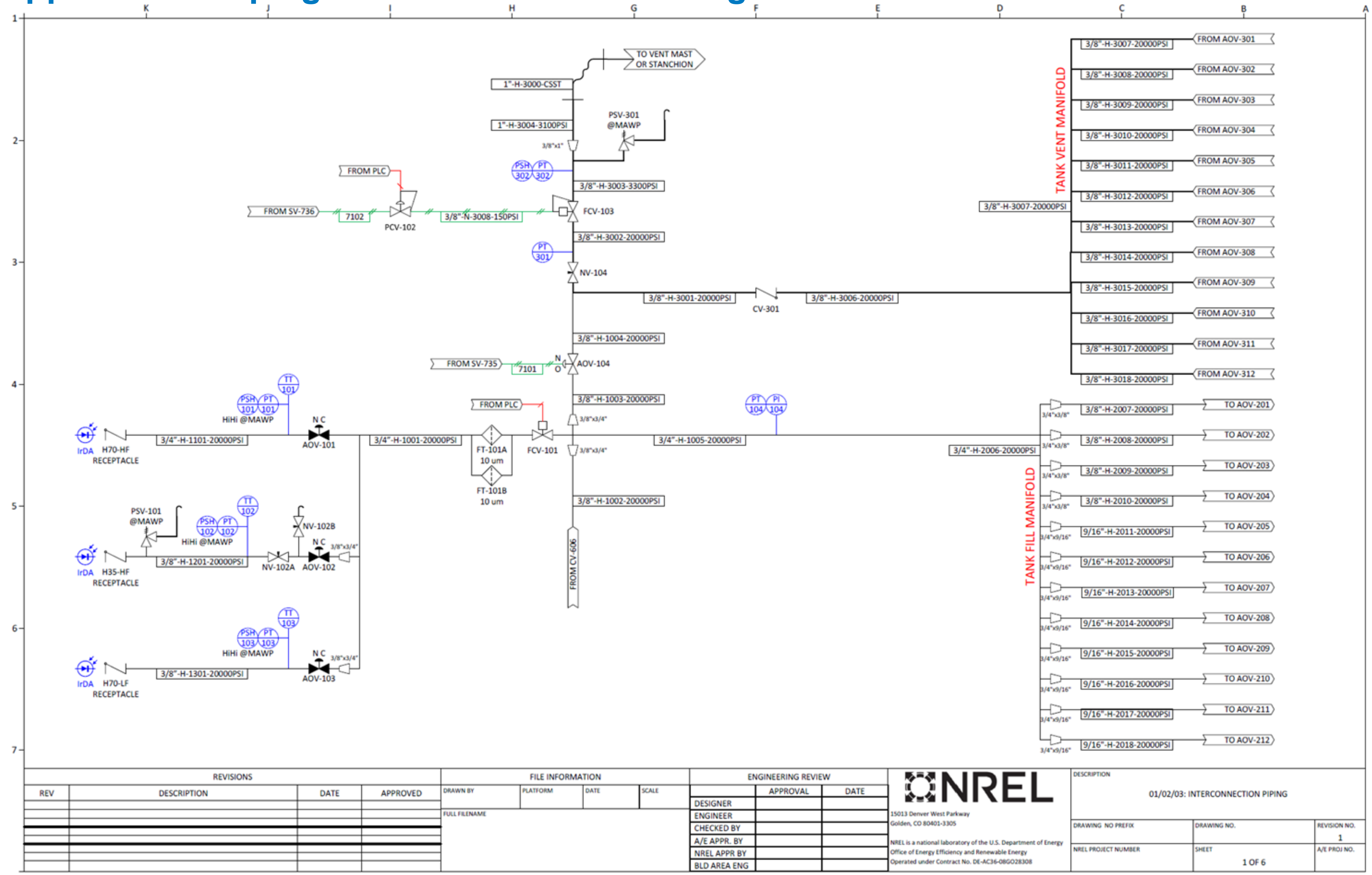
Included in Appendix D is an outline of an operation and maintenance (O&M) manual that should be provided with the HD HyStEP device. These sections are left uncompleted as their exact contents should be developed based on the as-built configuration of the HD HyStEP device and individual component manufacturer recommendations. This should also not be taken as an exhaustive collection of the required contents of the O&M manual. Best judgment should be used to determine the contents that are needed for the safe and practical use of the HD HyStEP device.

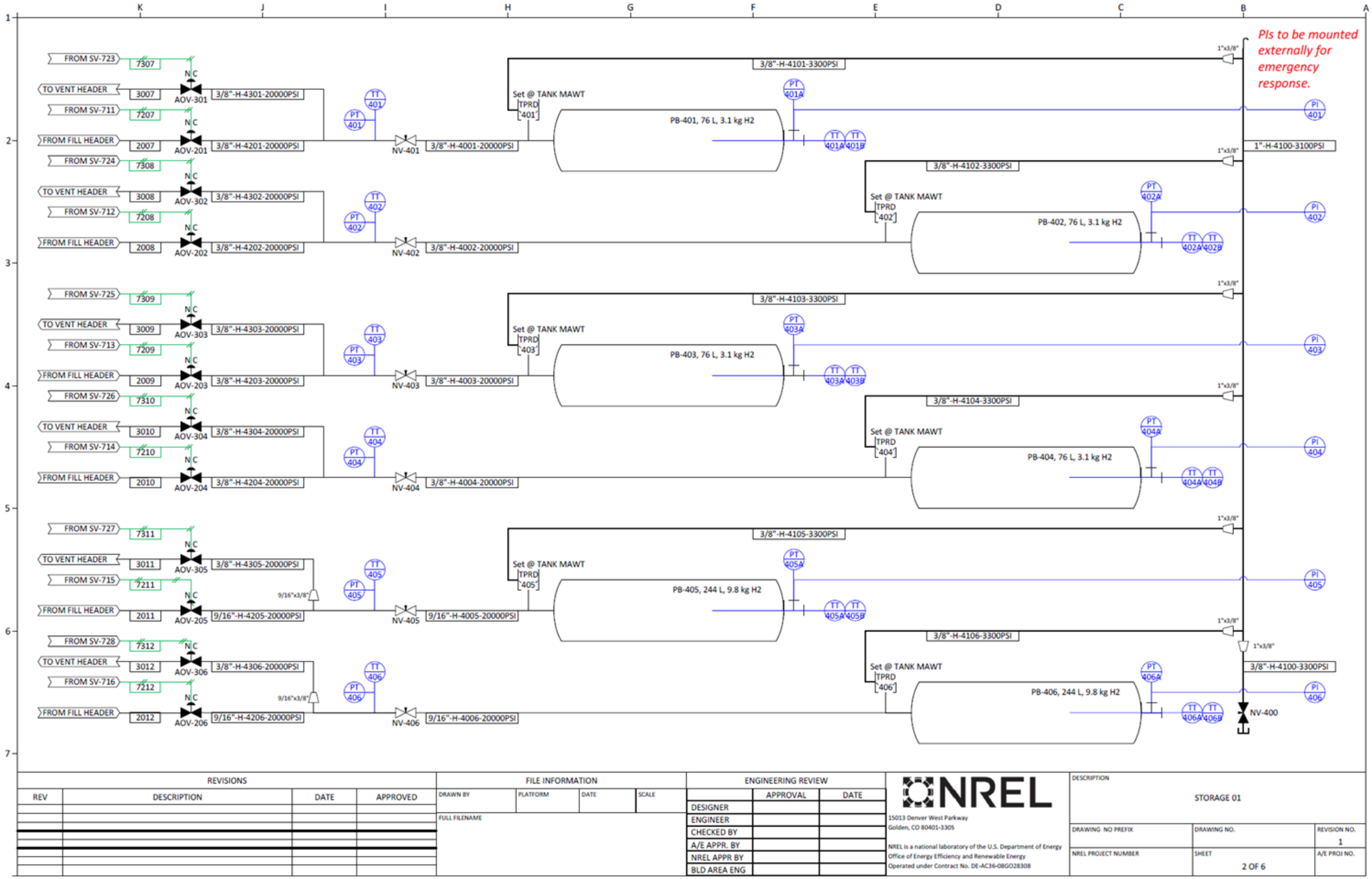
3.5 Process Hazard Analysis Outcomes

The information in Appendix E represents the output of a PHA that was conducted to evaluate the HD HyStEP device design outlined in the P&ID included in this report. In this PHA, the HD HyStEP device was broken into four nodes: (1) the interconnection piping, (2) the hydrogen storage vessels, (3) the inert gas system, and (4) the trailer that will enclose the other equipment. As each node was evaluated, several “deviations” from normal operation were considered. These deviations included conditions like “high pressure”, “reverse flow”, or “loss of containment” and the potential causes of those deviations were identified. If feasible causes of a given deviation were identified and were likely to cause a consequence with safety implications, those consequences were identified and recorded. For each consequence with safety implications, a series of safeguards were identified that could reduce the probability and/or severity of that consequence. After all relevant safeguards were identified, the team estimated the remaining severity and probability to determine the residual risk. If that residual risk was greater than the team deemed acceptable, additional recommended safeguards were identified before re-evaluating the probability and severity remaining after considering the recommendations. A final column includes “remarks”, which are not required in the final design but may be useful for the designer to consider.

Please note that although the team made their best reasonable effort to identify the risks associated with a potential HD HyStEP device, the designer and operator must use their best judgment regarding the safety of this device as it is designed, built, and operated. It is recommended that another PHA or similar safety review be conducted after the detailed design is near completion to identify any safety implications from design changes and/or any safety considerations that were missed during this original PHA. For example, many of the setpoints for safety devices like pressure relief valves, pressure switches, and thermal relief valves are left unspecified as they will depend on the precise components that are selected by the designer. These are safety critical devices, and so their selection and the impacts of that selection should be given the appropriate consideration to that criticality.

Appendix A. Piping and Instrumentation Diagram





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TO VENT HEADER
FROM SV-711
FROM FILL HEADER
FROM SV-724
FROM SV-712
FROM SV-725
FROM SV-713
FROM SV-726
FROM SV-714
FROM SV-727
FROM SV-715
FROM SV-728
FROM SV-716
AOV-301
AOV-201
AOV-302
AOV-202
AOV-303
AOV-203
AOV-304
AOV-204
AOV-305
AOV-205
AOV-306
AOV-206
NV-401
NV-402
NV-403
NV-404
NV-405
NV-406
NV-400
3/8"-H-4301-20000PSI
3/8"-H-4201-20000PSI
3/8"-H-4001-20000PSI
3/8"-H-4101-3300PSI
1"-H-4100-3100PSI
3/8"-H-4100-3300PSI
Set @ TANK MAWT
TPRD
PB-401, 76 L, 3.1 kg H2
PB-402, 76 L, 3.1 kg H2
PB-403, 76 L, 3.1 kg H2
PB-404, 76 L, 3.1 kg H2
PB-405, 244 L, 9.8 kg H2
PB-406, 244 L, 9.8 kg H2
9/16"-H-4205-20000PSI
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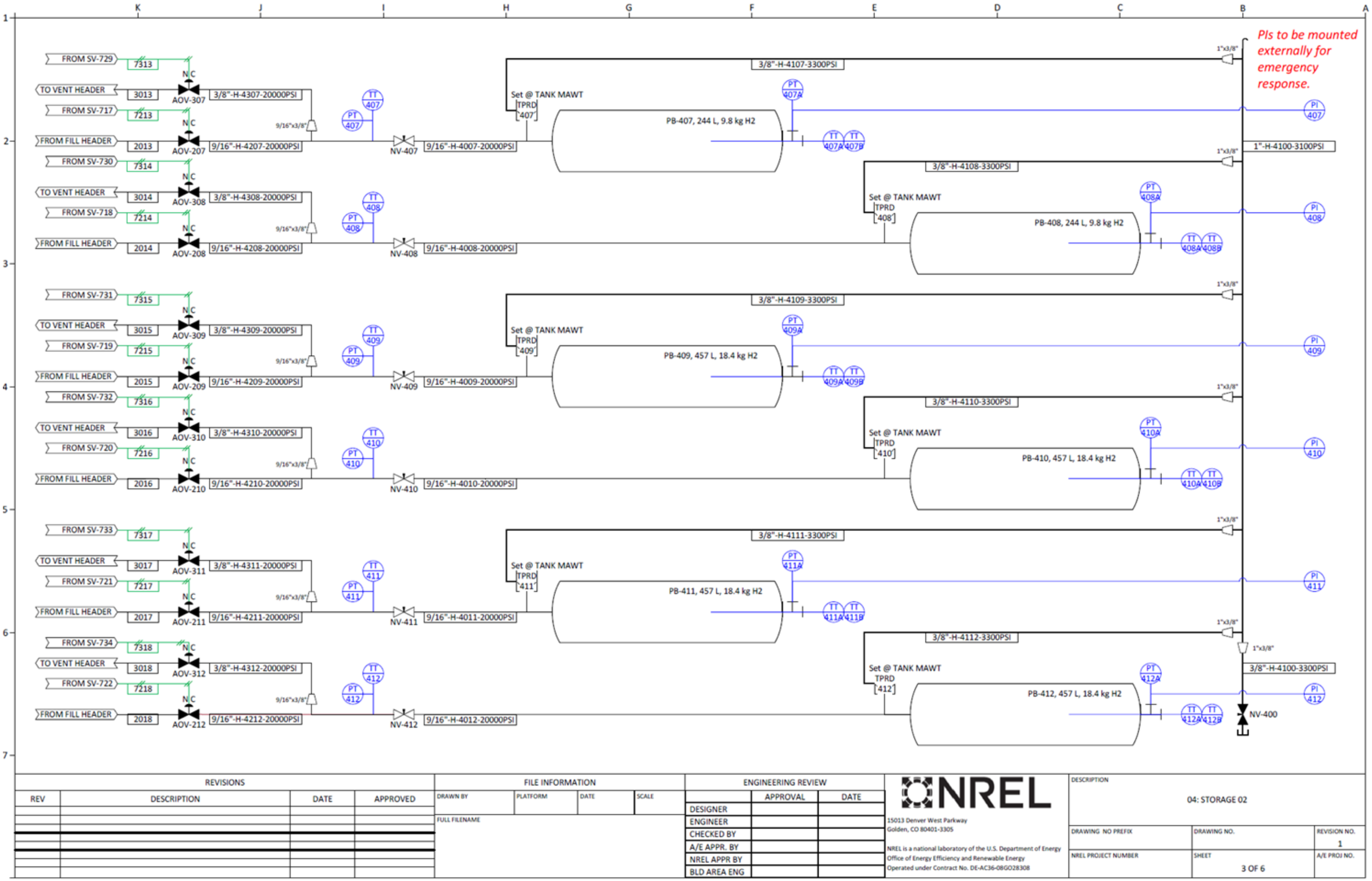
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FROM SV-717
FROM FILL HEADER
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AOV-207
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9/16"-H-4207-20000PSI
NV-407
9/16"-H-4007-20000PSI
3/8"-H-4107-3300PSI
Set @ TANK MAWT
TPRD 407
PB-407, 244 L, 9.8 kg H2
PB-408, 244 L, 9.8 kg H2
PB-409, 457 L, 18.4 kg H2
PB-410, 457 L, 18.4 kg H2
PB-411, 457 L, 18.4 kg H2
PB-412, 457 L, 18.4 kg H2
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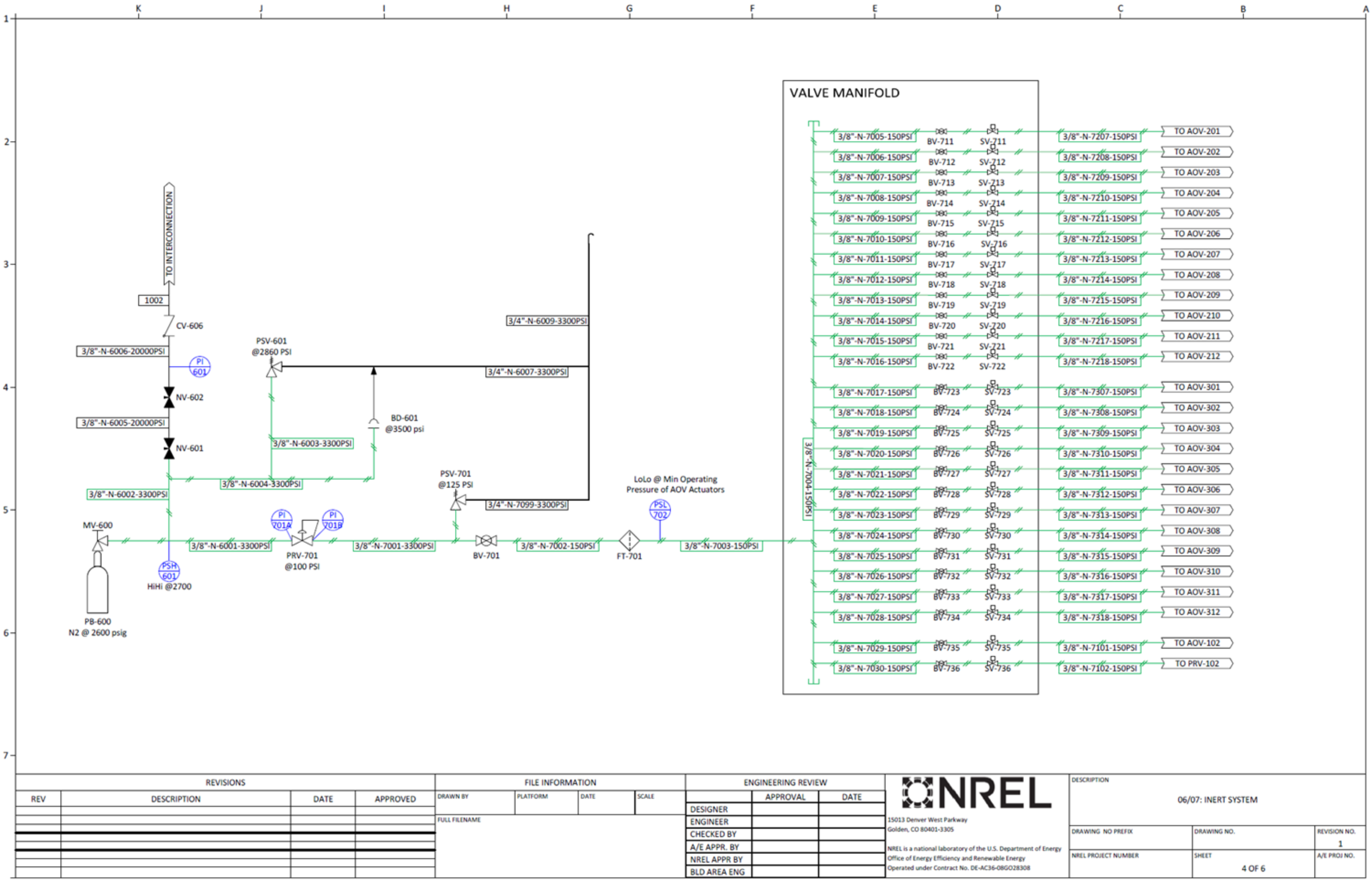
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3/8"-N-6002-3300PSI
MV-600
PB-600
N2 @ 2600 psig
PSH 601
HiHi @2700
3/8"-N-6001-3300PSI
3/8"-N-6004-3300PSI
3/8"-N-6003-3300PSI
PSV-601
@2860 PSI
BD-601
@3500 psi
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3/4"-N-6009-3300PSI
PI 701A
PI 701B
PRV-701
@100 PSI
3/8"-N-7001-3300PSI
PSV-701
@125 PSI
3/4"-N-7099-3300PSI
BV-701
3/8"-N-7002-150PSI
FT-701
LoLo @ Min Operating
Pressure of AOV Actuators
PSL 702
3/8"-N-7003-150PSI
3/8"-N-7004-150PSI
3/8"-N-7005-150PSI BV-711 SV-711 3/8"-N-7207-150PSI TO AOV-201
3/8"-N-7006-150PSI BV-712 SV-712 3/8"-N-7208-150PSI TO AOV-202
3/8"-N-7007-150PSI BV-713 SV-713 3/8"-N-7209-150PSI TO AOV-203
3/8"-N-7008-150PSI BV-714 SV-714 3/8"-N-7210-150PSI TO AOV-204
3/8"-N-7009-150PSI BV-715 SV-715 3/8"-N-7211-150PSI TO AOV-205
3/8"-N-7010-150PSI BV-716 SV-716 3/8"-N-7212-150PSI TO AOV-206
3/8"-N-7011-150PSI BV-717 SV-717 3/8"-N-7213-150PSI TO AOV-207
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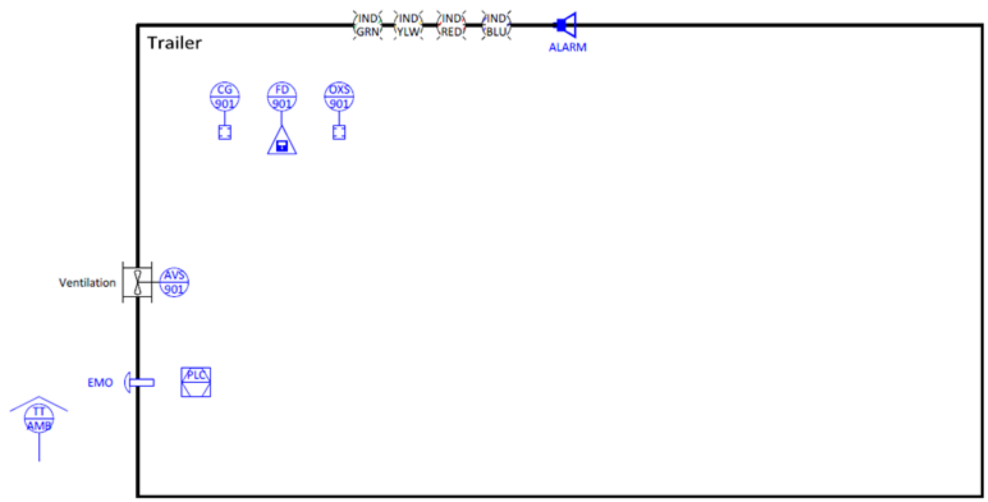

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B		Storage		
C	Compressor			Closed
D	Dispenser			
E	Electrolyzer			
F	Flow	Filter		
G	Chiller			
H	Hand			High
I		Indicator		
J	Power			
K				
L	Level			Low
M	Conductivity			Middle
N			Hydrogen	
O			Air	Open
P	Pressure			
Q				
R				
S			Switch	
T				
U				
V			Valve	
W			Water	
X				
Y				
Z	Position		Actuator	

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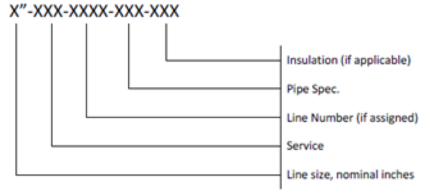

Line Service Identification codes
CCA-CLEAN COMPRESSED AIR
CHW-CHILLED WATER
H-HYDROGEN
N-NITROGEN

Insulation codes
F1- Soda-lime silicate glass (FOAMGLAS) 1" wall thickness
F2- Buna N Foam 3/8" thick

Pipe Specification Details

3000PSI	Min. pressure rating of 3,000 PSI 316/316L austenitic stainless steel tubing UNS S31600 (316), UNS S31603 (316L)
6500PSI	Min. pressure rating of 6,500 PSI 316/316L austenitic stainless steel tubing UNS S31600 (316), UNS S31603 (316L)
15000PSI	Min. pressure rating of 15,000 PSI 316/316L austenitic stainless steel tubing UNS S31600 (316), UNS S31603 (316L)
POLY	Polyamide (Nylon) pneumatic tubing, light and heat stabilized.
0.032CU	0.032" wall copper alloy 122 seamless tubing
SCH40	Schedule 40 iron pipe
CSST	Stainless steel braided hose.

VALVES

- BALL VALVE
- NEEDLE VALVE
- CHECK VALVE
- ELECTRICALLY DRIVEN VALVE
- N C ACTUATED VALVE, SPRING RETURN CLOSE
- PRESSURE REDUCING VALVE
- PRESSURE RELIEF

EQUIPMENT

- FILTER
- BURST DISC
- GAS BOTTLE
- HYDROGEN VESSEL
- FLAME DETECTOR
- WIRELESS INFRARED COMMUNICATIONS
- TPRD THERMAL RELIEF DEVICE

PROCESS AND SIGNAL LINES

- HYDROGEN PROCESS LINE
- INSTRUMENT CONNECTION LINE
- PNEUMATIC LINE
- ELECTRICAL SIGNAL LINE

REVISIONS			
REV	DESCRIPTION	DATE	APPROVED

FILE INFORMATION: DRAWN BY | PLATFORM | DATE | SCALE | FULL FILENAME

ENGINEERING REVIEW	APPROVAL	DATE
DESIGNER		
ENGINEER		
CHECKED BY		
A/E APPR. BY		
NREL APPR BY		
BLD AREA ENG		

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DESCRIPTION: LEGEND AND SPECIFICATIONS
DRAWING NO PREFIX | DRAWING NO. LEG | REVISION NO. 1
NREL PROJECT NUMBER | SHEET 6 OF 6 | A/E PROJ NO.

Appendix B. Bill of Materials

P&ID Tag No.	Qty.	Description	Example Specifications	MAWP (MPa)	MAWT (C)	MDMT (C)
		Interconnection and Vent System				
IrDA	3	Infrared wireless communications hardware	Capable of SAE J2799 Comms	N/A	N/A	N/A
PT-101	1	H70 HF Receptacle pressure transmitter	0-20,000 psig, +/- 0.1%FS	138	85	-40
TT-101	1	H70 HF Receptacle temperature sensor	Type T thermocouple	138	85	-40
PSH-101	1	H70 HF Receptacle pressure switch	Setpoint at the receptacle MAWP	138	85	-40
AOV-101	1	H70 HF Receptacle shutoff valve	N/C, 9/16", Cone & thread, Cv ~ 1.3	138	85	-40
PT-102	1	H35 HF Receptacle pressure transmitter	0-20,000 psig, +/- 0.1%FS	138	85	-40
TT-102	1	H35 HF Receptacle temperature sensor	Type T thermocouple	138	85	-40
PSH-102	1	H35 HF Receptacle pressure switch	Setpoint at the receptacle MAWP	138	85	-40
AOV-102	1	H35 HF Receptacle shutoff valve	N/C, 3/8", Cone & thread, Cv ~ 0.75	138	85	-40
NV-102A/B	2	H35 HF Receptacle manual block and vent valve	3/8", Cone & thread, Cv ~ 0.75	138	85	-40
PT-103	1	H70 LF Receptacle pressure transmitter	0-20,000 psig, +/- 0.1%FS	138	85	-40
TT-103	1	H70 LF Receptacle temperature sensor	Type T thermocouple	138	85	-40
PSH-103	1	H70 LF Receptacle pressure switch	Setpoint at the receptacle MAWP	138	85	-40
AOV-103	1	H70 LF Receptacle shutoff valve	N/C, 3/8", Cone & thread, Cv ~ 0.75	138	85	-40
FCV-101	1	dP adjustment valve	Electrically actuated, 9/16", Cone & thread, Cv ~ 1.3	138	85	-40
FT-101A/B	1	Inlet filter	5 um, 9/16" Cone & thread	138	85	-40
PI-104	1	Fueling manifold pressure indicator	0-20,000 psig, 316SS bourdon tube	138	85	-40
PT-104	1	Fueling manifold pressure transmitter	0-20,000 psig, +/- 0.1%FS	138	85	-40
CV-301	1	Vent manifold check valve	3/8" Cone & thread	138	85	-40
PT-301	1	Vent valve inlet pressure transmitter	0-20,000 psig, +/- 0.1%FS	138	85	-40
PT-302	1	Vent valve outlet pressure transmitter	0-5,000 psig, +/- 0.1%FS	34.5	85	-40
PSH-103	1	Vent valve outlet pressure switch	Setpoint at the downstream system MAWP	34.5	85	-40
AOV-104	1	Vent AOV	N/O, 3/8", Cone & thread, Cv ~ 0.75	87.5	85	-40
NV-104	1	Vent hand valve	3/8", Cone & thread, Cv ~ 0.75	87.5	85	-40
FCV-103	1	Vent flow control valve	Inlet pressures up to 15 ksi, Outlet up to 3000 psi	87.5	85	-40
PCV-102	1	Vent pressure regulator (as needed)	Control inert pressure from 0-150 psig, self venting	1	40	-20
-	2	Vent hose	H2 compatible, ~25 ft long each	20.7	85	-40
		Hydrogen Storage System				
PB-401 to 404	4	75 L Vessel	Type III/IV	87.5	85	-40
PB-405 to 408	4	250 L Vessel	Type III/IV	87.5	85	-40
PB-409 to 412	4	450 L Vessel	Type III/IV	87.5	85	-40
-	8	75 L Tank Plug	316 SS Plug with 3/8" Medium Pressure female connection	87.5	85	-40
-	8	250 L Tank Plug	316 SS Plug with 9/16" Medium Pressure female connection	87.5	85	-40
-	8	450 L Tank Plug	316 SS Plug with 9/16" Medium Pressure female connection	87.5	85	-40
AOV-201 to 204	4	Inlet AOV - 75 L	N/C, 3/8", Cone & thread, Cv ~ 0.75	138	85	-40
AOV-205 to 208	4	Inlet AOV - 250 L	N/C, 9/16", Cone & thread, Cv ~ 1.3	138	85	-40
AOV-209 to 212	4	Inlet AOV - 450 L	N/C, 9/16", Cone & thread, Cv ~ 1.3	138	85	-40

AOV-301 to 304	4	Vent AOV - 75 L	N/C, 3/8", Cone & thread, Cv ~ 0.75	138	85	-40
AOV-305 to 308	4	Vent AOV - 250 L	N/C, 3/8", Cone & thread, Cv ~ 0.75	138	85	-40
AOV-309 to 312	4	Vent AOV - 450 L	N/C, 3/8", Cone & thread, Cv ~ 0.75	138	85	-40
NV-401 to 404	4	Isolation valve - 75 L	3/8", Cone & thread	138	85	-40
NV-405 to 408	4	Isolation valve - 250 L	9/16", Cone & thread	138	85	-40
NV-409 to 412	4	Isolation valve - 450 L	9/16", Cone & thread	138	85	-40
PT-401 to 412	12	Vessel inlet pressure transmitter	0-20,000 psig, +/- 0.1%FS	138	85	-40
TT-401 to 412	12	Vessel inlet temperature sensor	Type T thermocouple	138	85	-40
PI-401 to 412	12	Vessel pressure transmitter	0-20,000 psig, 316SS bourdon tube, ≥4" face diameter	138	85	-40
PT-401A to 412A	12	Vessel pressure indicator	0-20,000 psig, H2 compatible	138	85	-40
TT-401A/B to 412A/B	12	Vessel temperature sensor (2 point)	Type T thermocouple	87.5	85	-40
NV-400	1	TPRD vent drain valve	Hydrogen compatible	22.8	40	-20
		Inert Gas System				
PB-600	1	Nitrogen bottle	Size 300			
CV-606	1	Purge connection check valve	3/8" Cone & thread	138	40	-40
NV-601 and 602	1	Purge connection hand valves	3/8" Cone & thread	138	40	-40
BD-601	1	Purge gas pressure burst disc	3500 psi Setpoint	22.8	40	-20
PSV-601	1	Purge gas pressure PSV	2860 psi Setpoint	22.8	40	-20
PSH-601	1	Purge gas pressure switch	2700 psi Setpoint	22.8	40	-20
PRV-701	1	Control gas pressure regulator	Inlet pressures up to 2700 psig, outlet up to 125 psig	22.8	40	-20
PSV-701	1	Control gas pressure PSV	125 psi Setpoint	1	40	-20
BV-701	1	Control gas shutoff valve	Ball valve	22.8	40	-20
FT-701	1	Control gas filter	10 um	1	40	-20
VALVE MANIFOLD	1	Control gas valve manifold	Normally closed, self venting solenoid valves with manual override	1	40	-20
		Trailer and Support Hardware				
PLC	1	Controls PLC	C1 D2 Grp B approved			
TT-AMB	1	Ambient TT				
CG-901	1	Enclosure CG	0-100% of LEL range, C1D1 Grp B approved			
FD-901	1	Enclosure flame detection	UV/IR Combined detector, C1D1 Grp B approved			
OXS-901	1	Oxygen concentration sensor	0-30% concentration range, C1D1 Grp B approved			
AVS-901	1	Air ventilation flow switch	C1D1 Grp B approved, fail open			
-	4	Indicator light(s)	Externally visible, indicating: warning, gas detection, and EMO			
EMO	1	External EMO button	Latching, N/O			
ALARM	1	Audible external alarm	C1 D2 Grp B approved			
-	1	Trailer	>4 Ton capacity, 8x8x24 or sized to fit equipment			

Appendix C. Cause and Effects Matrix

INPUT			TRIP					CSTOP RELAY	FCV-101	FCV-103	AOV-101	AOV-102	AOV-103	AOV-104	TANK FILL AOV-30[X]	TANK VENT AOV-20[X]	LIGHT IND-YLW	LIGHT IND-RED	LIGHT IND-BLU	VENTILATION	CONTROL POWER
ID (Input)	RANGE	DESCRIPTION	TYPE	NUMBER	TRIP LEVEL		DESCRIPTION	NO	FIP	NO	NC	NC	NC	NO	NC	NC	NO	NO	NO	NO	NO
E-Stop	On/Off	Station EPO / Loss of	E-Stop Station	1	All power lost	Fail	Station E-stop / power loss	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	Off	Off	Off	Off
EMO	On/Off	Onboard EMO button	EMO	3	DI-1 = 0 V	Fail	HDHS C-Stop	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
IRDA	-	Vehicle communications	Warning	5	Fault	Fail	Communications failure	On	No Action	No Action	No Action	No Action	No Action	No Action	No Action	No Action	On	Off	Off	On	On
TT-AMB	-270 - 400 C	Ambient Temperature	P-Stop	4	T > 50 C	HH	Ambient T hi-hi	On	No Action	No Action	Close	Close	Close	No Action	Close	Close	Off	On	Off	On	On
			Warning	5	T > 40 C	H	Ambient T hi	On	No Action	No Action	No Action	No Action	No Action	No Action	No Action	No Action	On	Off	Off	On	On
			Warning	5	T < _ C	L	Ambient T lo	On	No Action	No Action	No Action	No Action	No Action	No Action	No Action	No Action	On	Off	Off	On	On
			P-Stop	4	T < -40 C for _ seconds	LL	Ambient T lo-lo	On	No Action	No Action	Close	Close	Close	No Action	Close	Close	Off	On	Off	On	On
			P-Stop	4	Out of range	OoR	Receptacle T failure	On	No Action	No Action	Close	Close	Close	No Action	Close	Close	Off	On	Off	On	On
CG-901	0-100% LEL	Trailer combustible gas	Gas Detect	6	CG > 25 % LEL	HH	CG hi-hi relay	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	On	On	Off
			P-Stop	4	CG > 10 % LEL	H	CG hi relay	On	No Action	No Action	Close	Close	Close	No Action	Close	Close	Off	On	Off	On	On
			EMO	3	Out of range	OoR	CG sensor failure	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
			EMO	3	Fault	Open	CG fault relay	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
FD-901	-	Trailer flame detection	Fire Alarm	2	Fire detected	Open	Trailer FD fire alarm relay	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	Off	Off
			EMO	3	Fault	Open	Trailer FD fault relay	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
OXS-901	-	Trailer oxygen sensor	Gas Detect	6	Oxygen < 19.5%	Open	Trailer oxygen lo-lo	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	On	On	Off
			Gas Detect	6	Oxygen > 23.5%	Open	Trailer oxygen hi-hi	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	On	On	Off
			EMO	3	Out of range	OoR	Trailer oxygen sensor failure	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
			EMO	3	Fault	Open	Trailer oxygen sensor fault relay	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
AVS-901	-	Trailer ventilation switch	EMO	3	Loss of ventilation flow	Open	Trailer vent. switch relay	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
			EMO	3	Fault	Open	Trailer vent. switch fault relay	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
FCV-101	4-20 mA	Flow valve positioner	P-Stop	4	Out of range	OoR	Positioner failure	On	No Action	No Action	Close	Close	Close	No Action	Close	Close	Off	On	Off	On	On
PSH-101	0 - 24 V	H70-HF Receptacle pressure switch	EMO	3	P > 87.5 MPa	Open	Receptacle pressure switch relay	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
			EMO	3	Fault	Open	Receptacle pressure switch failure	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
PSH-102	0 - 24 V	H35-HF Receptacle pressure switch	EMO	3	P > 45 MPa	Open	Receptacle pressure switch relay	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
			EMO	3	Fault	Open	Receptacle pressure switch failure	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
PSH-103	0 - 24 V	H70-LF Receptacle pressure switch	EMO	3	P > 87.5 MPa	Open	Receptacle pressure switch relay	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
			EMO	3	Fault	Open	Receptacle pressure switch failure	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
PT-101	0 - 100 MPa	H70-HF Receptacle pressure	EMO	3	P > 87.5 MPa	HH	Receptacle P hi-hi	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
			Warning	5	P > _ MPa	H	Receptacle P hi	On	No Action	No Action	No Action	No Action	No Action	No Action	No Action	No Action	On	Off	Off	On	On
			EMO	3	Out of range	OoR	Receptacle P failure	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
TT-101	-270 - 400 C	H70-HF Receptacle temperature	EMO	3	T > 80 C	HH	Receptacle T hi-hi	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
			Warning	5	T > 50 C	H	Receptacle T hi	On	No Action	No Action	No Action	No Action	No Action	No Action	No Action	No Action	On	Off	Off	On	On
			Warning	5	T < -40 C for < 5 s	L	Receptacle T lo	On	No Action	No Action	No Action	No Action	No Action	No Action	No Action	No Action	On	Off	Off	On	On
			EMO	3	T < -40 C for > 5 s	LL	Receptacle T lo-lo	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
			EMO	3	Out of range	OoR	Receptacle T failure	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
PT-102	0 - 100 MPa	H35-HF Fill header pressure	EMO	3	P > 45 MPa	HH	Receptacle P hi-hi	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
			Warning	5	P > _ MPa	H	Receptacle P hi	On	No Action	No Action	No Action	No Action	No Action	No Action	No Action	No Action	On	Off	Off	On	On
			EMO	3	Out of range	OoR	Receptacle P failure	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
TT-102	-270 - 400 C	H35-HF Receptacle temperature	EMO	3	T > 80 C	HH	Receptacle T hi-hi	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
			Warning	5	T > 50 C	H	Receptacle T hi	On	No Action	No Action	No Action	No Action	No Action	No Action	No Action	No Action	On	Off	Off	On	On
			Warning	5	T < -40 C for < 5 s	L	Receptacle T lo	On	No Action	No Action	No Action	No Action	No Action	No Action	No Action	No Action	On	Off	Off	On	On
			EMO	3	T < -40 C for > 5 s	LL	Receptacle T lo-lo	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
			EMO	3	Out of range	OoR	Receptacle T failure	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
PT-103	0 - 100 MPa	H70-LF Receptacle pressure	EMO	3	P > 87.5 MPa	HH	Receptacle P hi-hi	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
			Warning	5	P > _ MPa	H	Receptacle P hi	On	No Action	No Action	No Action	No Action	No Action	No Action	No Action	No Action	On	Off	Off	On	On
			EMO	3	Out of range	OoR	Receptacle P failure	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
TT-103	-270 - 400 C	H70-LF Receptacle temperature	P-Stop	4	T > 80 C	HH	Receptacle T hi-hi	On	No Action	No Action	Close	Close	Close	No Action	Close	Close	Off	On	Off	On	On
			Warning	5	T > 50 C	H	Receptacle T hi	On	No Action	No Action	No Action	No Action	No Action	No Action	No Action	No Action	On	Off	Off	On	On
			Warning	5	T < -40 C for < 5 s	L	Receptacle T lo	On	No Action	No Action	No Action	No Action	No Action	No Action	No Action	No Action	On	Off	Off	On	On
			EMO	3	T < -40 C for > 5 s	LL	Receptacle T lo-lo	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
			EMO	3	Out of range	OoR	Receptacle T failure	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
PT-104	0 - 100 MPa	Fill header pressure	EMO	3	P > 87.5 MPa	HH	Fill header P hi-hi	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
			Warning	5	P > _ MPa	H	Fill header P hi	On	No Action	No Action	No Action	No Action	No Action	No Action	No Action	No Action	On	Off	Off	On	On
			EMO	3	Out of range	OoR	Fill header P failure	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
PT-40[X]	0 - 100 MPa	PB-40[X]: Vessel Inlet Pressure	EMO	3	P > 87.5 MPa	HH	Vessel inlet P hi-hi	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
			Warning	5	P > _ MPa	H	Vessel inlet P hi	On	No Action	No Action	No Action	No Action	No Action	No Action	No Action	No Action	On	Off	Off	On	On
			EMO	3	P < 0.5 MPa	LL	Vessel inlet P lo-lo	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
			EMO	3	Out of range	OoR	Vessel inlet P failure	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
			P-Stop	4	T > 85 C	HH	Vessel T hi-hi	On	No Action	No Action	Close	Close	Close	No Action	Close	Close	Off	On	Off	On	On
			Warning	5	T > _ C	H	Vessel T hi	On	No Action	No Action	No Action	No Action	No Action	No Action	No Action	No Action	On	Off	Off	On	On

TT-40[X]	-270 - 400 C	PB-40[X]: Vessel Inlet Temperature	Warning	5	T < _ C	L	Vessel T lo	On	No Action	No Action	No Action	No Action	No Action	No Action	No Action	No Action	On	Off	Off	On	On
			EMO	3	T < -40 C	LL	Vessel T lo-lo	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
			EMO	3	|TA-TB| > 10 C	OoR	Vessel T disagreement	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
			EMO	3	Out of range	OoR	Vessel T failure	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
TT-40[X]A/B	-270 - 400 C	PB-40[X]: Vessel Bulk Temperature	P-Stop	4	T > 85 C	HH	Vessel inlet T hi-hi	On	No Action	No Action	Close	Close	Close	No Action	Close	Close	Off	On	Off	On	On
			Warning	5	T > _ C	H	Vessel inlet T hi	On	No Action	No Action	No Action	No Action	No Action	No Action	No Action	No Action	On	Off	Off	On	On
			Warning	5	T < _ C	L	Vessel inlet T lo	On	No Action	No Action	No Action	No Action	No Action	No Action	No Action	No Action	On	Off	Off	On	On
			EMO	3	T < -40 C	LL	Vessel inlet T lo-lo	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
			EMO	3	Out of range	OoR	Vessel inlet T failure	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
PT-40[X]A	0 - 100 MPa	PB-40[X]: Vessel Pressure	EMO	3	P > 87.5 MPa	HH	Vessel P hi-hi	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
			Warning	5	P > _ MPa	H	Vessel P hi	On	No Action	No Action	No Action	No Action	No Action	No Action	No Action	No Action	On	Off	Off	On	On
			EMO	3	P < 0.5 MPa	LL	Vessel P lo-lo	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
			P-Stop	4	dP/dt Out of corridor	OoR	Vessel pressure ramp rate hi/lo	On	No Action	No Action	Close	Close	Close	No Action	Close	Close	Off	On	Off	On	On
			P-Stop	4	dP/dt < OEM Max Vent	LL	Vessel vent rate too high	On	No Action	No Action	Close	Close	Close	No Action	Close	Close	Off	On	Off	On	On
			EMO	3	Out of range	OoR	Vessel P failure	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
PSH-601	0 - 24 V	Inert Gas Supply Pressure Switch	EMO	3	P > 22.75 MPa	HH	Inert Gas P switch high high	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
			EMO	3	Fault	Open	Inert Gas P switch failure	Off	No Action	Open	Close	Close	Close	Open	Close	Close	Off	On	Off	On	Off
PSL-702	0 - 24 V	Inert Gas Control Pressure Switch	P-Stop	4	P < P_min from valve OEM	LL	Control P switch low low	On	No Action	No Action	Close	Close	Close	No Action	Close	Close	Off	On	Off	On	On
			P-Stop	4	Fault	Open	Control P switch failure	On	No Action	No Action	Close	Close	Close	No Action	Close	Close	Off	On	Off	On	On

Appendix D. Operation and Maintenance Manual Outline

Startup Procedures

Prior to First Time Use

Validating Alarms

[Checklist and/or instructions for validating the functionality of safety alarms on the HD HyStEP device before initial use. This may be accomplished via "spoofing" of variables when it is not safe and/or practical to achieve actual alarm conditions. In this case, it is critical to verify that instruments are properly installed and calibrated.]

Initial Leak Checks and Purging

[Instructions for leak checking and then purging the HD HyStEP device tubing and vessels. These instructions should include the use of an inert gas for primary leak checking, the use of helium or another inert tracer gas for small leak detection, and the use of torque checking compound to indicate that connections have been torqued appropriately as is applicable.]

Prior to Each Use

Site Inspection

[Checklist/instructions for verifying that the site and HD HyStEP device are suitable for the planned tests. Consideration should be given to overhead clearances at the fueling location and the approach to that location. The location and accessibility of station-side emergency response devices (e.g., EMO buttons, fire detectors, etc.) should additionally be considered when identifying appropriate placement of the HD HyStEP and remote vent mast (when used).]

Startup Leak Checks

[Instructions to verify that the system has not developed a major leak after transport to a station. Inert-gas leak checks should be conducted of all piping up to the isolation valves of the hydrogen storage vessels. Include instructions for the inspection of connections that cannot be feasibly leak checked (i.e., those connections between each tank and its associated isolation valve).]

Connecting to Power

[Instructions for connecting the HD HyStEP device to site electrical power.]

Making Vent Connections

Erecting Stand-Alone Vent Mast

[Instructions for siting, assembling, and connecting the stand-alone vent mast. Include requirements for drive-overs or similar protections for hoses that cross areas where vehicles and/or people may traverse.]

Connecting to Station Vent

[Instructions for connecting to the station vent if it is deemed safe and feasible by the station owner and HD HyStEP device operators. Include criteria for evaluating the suitability of this vent. Some of those criteria will include the length, diameter, and routing of the station vent.]

Regular Use Procedures

Typical Test Matrix for a Station

[Include a table of the tests that will be conducted at a given station.]

Configuring the Device for a Specific Test

PLC Configuration

[Instructions for selecting the correct test parameters using the onboard PLC interface to prepare the HD HyStEP for a test.]

Vessel Configuration

[Configuring the vessels for the correct TVL and CHSS category.]

Nozzle Selection and Configuration

[Selecting the nozzle needed for a specific test. This should include special attention to ensuring that the H35 HF nozzle is isolated in cases that it is not being utilized.]

Communications Setup

[Instructions for setting up and/or enabling the IRDA communications for a communications fill.]

Depressurization Procedure

[Instructions for safely venting the vessels after the completion of a test. Note that this may need to be conducted without electrical power, so the procedure should allow for this operation in the case that there is no electrical power available.]

Returning Hydrogen to a Fueling Station

[If it is deemed feasible and agreed to by both the fueling station and HD HyStEP device owners/operating entities, the system may be configured to return hydrogen gas to a fueling station via the station's hydrogen delivery stanchion. If this is an expected operation, this manual should include instructions for safely conducting this defueling procedure.]

Shutdown Procedures

[Instructions for shutting down the HD HyStEP device and making it safe to transport. This will include (in no specific order): venting the hydrogen and inert systems to below 30 psig, removing electrical power, disassembling and storing the remote vent mast, and otherwise readying the trailer for transport.]

Troubleshooting

[Instructions for recovering from likely / common issues that are expected to occur during the lifetime of the device.]

List of Faults and Their Meaning

[All faults that may appear to the user should be described here. If any of the external lights are activated in the case of this fault, that should be indicated here in addition to any messages that appear on the HMI.]

Repairing the PLC Software

[Instructions for repairing the PLC in case of an error/data loss/other software corruption.]

Recovering from Faults

[Include a list of alarms that the user may encounter, what they indicate, and what actions should be taken in case of a specific fault. Some likely faults that should be addressed are listed below.]

- Protocol Faults
    - [Recovery procedures to reset in the case that a dispenser does not complete a fill because of a failure to meet protocol requirements.]
- IrDA Communication Faults
    - [Instructions for addressing failures of IrDA communications between the HD HyStEP device and the dispenser.]
- High Pressure Faults
    - [This should include procedures for recovering from a high pressure detection at each of the sensing points. Special consideration should be given to the case of a high pressure detection on the inert gas system that could indicate that high-pressure hydrogen has leaked backwards into this system.]
- Low Pressure Faults
    - [This will be most important in the case of low pressure in the control gas system that actuates the valves.]
- Other Faults:
    - Combustible Gas Alarm
    - Loss of Ventilation
    - Low Oxygen Inside Trailer
    - High Temperature
    - Low Temperature
    - High Pressure
    - Instrument Failure
    - Loss of Power

Emergency Response

[Emergency response procedures.]

External Fire Response

[Explanation of HyStEP-specific actions/response in case of a fire outside of the device (e.g., at the nearby station).]

Internal Fire Response

[Explanation of HD HyStEP-specific actions/response in case of a fire inside of the HD HyStEP device trailer.]

Major Leak Response

[Explanation of HD HyStEP-specific actions/response in case of a major leak onboard the HD HyStEP device.]

Power-Loss Response

[Explanation of HD HyStEP-specific actions/response in case of a total loss of power to the HD HyStEP device during operation. This should consider the steps needed to determine whether the system is safe to re-energize and/or if the trailer is safe to be opened and entered for troubleshooting. Special care should be taken as this condition will include a loss of ventilation and gas monitoring.]

Data and Controls

[Information about the control interface and the data that the PLC will create.]

HMI Screens

[Descriptions of HMI screen inputs/outputs with screenshots.]

Data Fields

[Table or list of each data field with a brief description of the fields.]

Communicating with the PLC

[Instructions for communicating with the PLC via an ethernet, USB, etc. connection if available. This may be required for troubleshooting and/or for updating software as needed.]

Downloading Data from the PLC

[Instructions for downloading data from the PLC to an external device.]

Maintenance Procedures

Purging Procedures

[Instructions for purging different sections of tubing and/or vessels for maintenance activities. Note that there may not be electrical power during this activity, so procedures should be developed such that the system can be safely purged in this condition.]

LOTO Procedures

[Instructions for locking out different sections of the HD HyStEP device to perform maintenance.]

Calibration Schedule

[List of items requiring periodic calibration and the specific period for each (e.g., PSVs, PTs, etc.).]

Other Important Information

[This section will include other information that should be included with the device.]

Piping and Instrumentation Diagram (P&ID)

[Include graphical P&ID of the HD HyStEP device as built.]

Electrical Schematic

[Include schematics of the power and controls wiring as built.]

Cause and Effects Matrix

[Include a cause and effect matrix (see example) that illustrates the expected actions to be taken by the controls system in the case of reaching each alarm condition.]

System BOM

[List / table of components that were used to construct the HD HyStEP device. Include important specifications as relevant to each component (e.g., part number, power requirement, pressure, temperature, etc.).]

Tooling and Spares List

[List of items to be kept onboard for maintenance (e.g., special tooling, LOTO devices, rebuild kits, etc.).]

Appendix E. Process Hazard Analysis Outcomes

General

Administration

Facility Information

Company: NREL

Business Unit: 5700

Facility: Heavy Duty Hydrogen Station Equipment Performance Mobile Skid

Project ID: HDHS

Project Name: HyStEP

Study Duration

Start Date: 2024-01-29

End Date: 2024-04-01

Procedure

Procedure Document Name:

Procedure Document Number:

Procedure Document Revision:

Scope and Objective

Scope: The Heavy Duty HyStEP mobile trailer design will be deployed to hydrogen stations in California to validate SAE fueling protocols in development for heavy duty hydrogen fueling. This PHA identified hazards and controls associated with the HDHS using a HAZOP (hazard and operability study) PHA (process hazard analysis) methodology.

Objective: The primary objective of this study was to identify possible process hazard scenarios (credible cause and consequence combinations) that may exist or be introduced with the operation of the HDHS at Heavy Duty Flow Hydrogen Dispensing Stations in the State of California. If a credible hazard could occur, the adequacy of the existing safeguards is weighed against the probability and severity of the scenario to determine whether modifications to the process or additional safeguards are necessary.

This study was conducted using the "HAZOP" analysis technique for the gaseous hydrogen integration. A HAZOP study is a structured and systematic examination of a planned or existing process or operation. Hazards and potential operating problems are then sought by considering possible deviations from the design intention of the section under review. The design intention is a word picture of what should be happening and should contain all of the key parameters that will be explored during the study. The study also includes a statement of the intended operating range (envelope). For those deviations where the team can suggest a cause, the consequences are estimated using the team's experience and existing safeguards are taken into account. Where the team considers the risk to be nontrivial or where an aspect requires further investigation, a formal record is generated to allow the problem to be followed up outside the meeting. The team then moves on with the analysis.

Drawings / References

Document Number	Document Title	Rev.	File Attachment/Path	Comment	Place(s) Used
HDHS- pg 1 through 7	HDHyStEP_032524 PHA_Update	3/25/2024			

Study Nodes

Nodes	Type	Design Conditions/Parameters	Drawings / References	Equipment		Comment
				Tag	Description	
1. Interconnecting Piping		MAOP: 12690 psig (87.5 MPa) MPa, MAWP: 13959 psig (96.2 MPa); MAOT: -40C min, 85C max; Flow: 20 kg/min peak, 10 kg/min avg Hydrogen			-Assume Station pressures providing H2 to HD HyStEP are protected to NFPA 2 HD, ASME and CSA code -HD HyStEP protects itself, not expected that station fueling protocols will shutdown fueling	
2. H2 Storage Tanks		MAOP: xx MPa, MAWP: xx MPa; MAOT: -40C min, 85C max; Flow: 20 kg/min peak, 10 kg/min avg Hydrogen			-Omit OTV and added TPRDs separately from devices, two independent thermocouples outside of the bulk gas flow	
3. Inert Gas System		Purge N2: MAOP/MAWP: cylinder pressure 2500 psig; MAOT ambient temperatures Instrumentation N2: MAOP/MAWP: 125 psig; MAOT: ambient temperatures			P-stop: shutting down all inlet and outlet valves (valves AOV-301 through AOV-312 and AOV-201 through AOV-212) and AOV-101 through AOV-103. Vent AOV-104 opens to vent the fill header.	
4. Trailer						

HAZOP Deviations

Node: 1. Interconnecting Piping

Drawings / References:

Design Conditions/Parameters: MAOP: 12690 psig (87.5 MPa) MPa, MAWP: 13959 psig (96.2 MPa);
MAOT: -40C min, 85C max;
Flow: 20 kg/min peak, 10 kg/min avg
Hydrogen

Deviations	Parameter	Guide Word
1. High Flow		
2. No/Low Flow		
3. Misdirected Flow		
4. Reverse Flow		
5. Loss of Containment		
6. High Pressure		
7. Low Pressure		
8. High Temperature		
9. Low Temperature		
10. High Level		
11. Low/No Level		
12. Other than Composition - Contamination, Impurities, Air		
13. Other than Sequence - Wrong action in operation, Operation not completed		
14. Control System Failure		

Deviations	Parameter	Guide Word
15. Power Failure		
16. Operator Interaction		

Node: 2. H2 Storage Tanks
Drawings / References:

Design Conditions/Parameters: MAOP: xx MPa, MAWP: xx MPa;
MAOT: -40C min, 85C max;
Flow: 20 kg/min peak, 10 kg/min avg
Hydrogen

Deviations	Parameter	Guide Word
1. High Flow		
2. No/Low Flow		
3. Misdirected Flow		
4. Reverse Flow		
5. Loss of Containment		
6. High Pressure		
7. Low Pressure		
8. High Temperature		
9. Low Temperature		
10. High Level		
11. Low/No Level		
12. Other than Composition - Contamination, Impurities, Air		
13. Other than Sequence - Wrong action in operation, Operation not completed		
14. Control System Failure		
15. Power Failure		

Deviations	Parameter	Guide Word
16. Operator Interaction		

Node: 3. Inert Gas System

Drawings / References:

Design Conditions/Parameters: Purge N2: MAOP/MAWP: cylinder pressure 2500 psig; MAOT ambient temperatures
Instrumentation N2: MAOP/MAWP: 125 psig; MAOT: ambient temperatures

Deviations	Parameter	Guide Word
1. High Flow		
2. No/Low Flow		
3. Misdirected Flow		
4. Reverse Flow		
5. Loss of Containment		
6. High Pressure		
7. Low/Vacuum Pressure		
8. High Temperature		
9. Low/Cryogenic Temperature		
10. High Level		
11. Low/No Level		
12. Other than Composition - Contamination, Impurities, Air		
13. Other than Sequence - Wrong action in operation, Operation not completed		
14. Control System Failure		
15. Power Failure		
16. Operator Interaction		

Node: 4. Trailer

Drawings / References:

Design Conditions/Parameters:

Deviations	Parameter	Guide Word
1. Transportation - Vehicle Incident		
2. External Fire		
3. Operator Interaction		
4. Public Interaction		

HAZOP Study Worksheet

Node: 1. Interconnecting Piping

Drawings / References:

Design Conditions/Parameters: MAOP: 12690 psig (87.5 MPa) MPa, MAWP: 13959 psig (96.2 MPa);
MAOT: -40C min, 85C max;
Flow: 20 kg/min peak, 10 kg/min avg
Hydrogen

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk			HAZOP Recommendations		Residual Risk			Remarks
							S	P	RR	Proposed Recommendations	Responsibility	S	P	RR	
1. High Flow	1. Dispenser providing flow outside of pressure ramp rate (>20 kg/min)	1. Overtemperature of tanks - See Node 2	S												
	2. Too high of setpoint FCV-101 dispensing flow control valve	1. Operational impacts only in this node	BI												
	3. Too high setpoint FCV-103 venting flow control valve	1. High flow leads to high pressure. See High Pressure below	S												
2. No/Low Flow	1. Plugged filter (FT-101A/B), FCV-101 flow control valve closed/setpoint too low, NV-101, NV-102 or NV-103 closed when should be opened for fill	1. Fall outside of pressure ramp rate in fueling protocol. Operational only, no safety consequence.	BI												
	2. FCV-103 venting flow control valve in closed/too low	1. Not able to vent HD HyStEP vessels. H2 in system.	S												

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk			HAZOP Recommendations		Residual Risk			Remarks
							S	P	RR	Proposed Recommendations	Responsibility	S	P	RR	
	setpoint, downstream pressure sensor PT-301 or PT-302, loss of N2 or power to control FCV-103	See Loss of Control System and See Loss of Power													
3. Misdirected Flow	1. Operator error - H35 NV-102 left open during H70 filling	1. Overpressure H35 receptacle check valve leading to loss of H2, potential for ignition and fire	S	3. PSV-101 H35 relief valve set @ 6500 psig (45 MPa) sized for flow of H70 HF fill, vented to CGA G-5.5 emergency vent stack		E	I	E	Low						1. Consider if H35 hand valve NV-102 can be downsized, i.e. 9/16" line to help restrict flow of H70 high flow to PSV-101 (i.e. 20 kg/min)
				4. PSH-102/PT-102 high pressure initiates emergency stop fill (EMO) to HD HyStEP. Action AOV-201 through 212 close and initiates pressure to spike outside of the pressure ramp window.		E									2. Consider replacing PSV-101 with burst disk PSD-101
				1. Fueling protocol dictates pressure ramp rates and if falls outside window above the pressure ramp rate high-high, then the station should shut down the fill within 5 seconds.		E									3. Consider installing check valve directly upstream or downstream of NV-103.
				5. Attended operations during fills by trained and qualified operators		T									4. Install double block and bleed for H35 HF NV-102A/NV-102B - look at air introduction if only one valve with bleed, with one being an AOV.
				6. Station EPO Button		P									5. Protocol clarification on H35 HF fill to determine Cv of NV-102

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk S	Current Risk P	Current Risk RR	HAZOP Recommendations: Proposed Recommendations	HAZOP Recommendations: Responsibility	Residual Risk S	Residual Risk P	Residual Risk RR	Remarks
				7. Stations: UV/IR flame detection EPOs station and trips fire alarm		E									6. Replace needle valves (NV-101, 102 and 103) with Air Operated valves with hard interlocks (exclusive Or) between the three receptacles 7. Check CSA to make sure station owners do not have concern with sudden shutoff of HD HyStEP - concern with hose pressure spike
		2. High flow of -40C H2 through PSV-101 potential for PSV vent to flare and high levels of H2 in vicinity. Potential exceeding LEL of H2 (overhead awnings may be part of station design), H2 ignition, and fire	S	8. Hazard Classified area Class 1, Div 2 (15 feet for dispenser) with immediate vent path Class 1, Div 1 (5 feet)		E	I	E	Low						
				10. Emergency PSV vent designed to CGA G-5.5		E									
				4. PSH-102/PT-102 high pressure initiates emergency stop fill (EMO) to HD HyStEP. Action AOV-201 through 212 close and initiates pressure to spike outside of the pressure ramp window.		E									
				1. Fueling protocol dictates pressure ramp rates and if falls outside window above the pressure ramp rate high-high, then the station should shut down the fill within 5 seconds.		E									
				5. Attended operations during fills by trained and qualified operators		T									

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk			HAZOP Recommendations		Residual Risk			Remarks
							S	P	RR	Proposed Recommendations	Responsibility	S	P	RR	
				6. Station EPO Button		P									
				7. Stations: UV/IR flame detection EPOs station and trips fire alarm		E									
				9. Station dispenser fixed gas detector for LEL-H2 at 50% of LEL initiates Station EPO		E									
				11. NFPA 2 setback for emergency ventilation met		E									
	2. Operator error - Wrong H70 receptacle hand valve opened NV-101 (HF) or NV-103 (LF), during an H70 fill	1. Operational only, no safety consequence	BI												
		2. Potential to exceed cycle life of receptacle. Lead to leak in receptacle, H2 release, ignition and fire. **Unknown if this is an issue**	S	12. Receptacle cycle life counter implemented (PT-101, PT-102 and PT-103) and receptacle changed out when cycle life exceeded.		P	I			1. Evaluate if receptacle life is a concern and add receptacle cycle life counter within control system via PT-101, PT-102 and PT-103					
				8. Hazard Classified area Class 1, Div 2 (15 feet for dispenser) with immediate vent path Class 1, Div 1 (5 feet)		E				2. Recommend operators PPE to include personal CG/O2 detector and FR clothing and FR gloves when dispensing or conducting maintenance of HDHS.					
				9. Station dispenser fixed gas detector for LEL-H2 at 50% of LEL initiates Station EPO		E				3. Implement alarm on PT-101 or PT-103 high pressure alarm					
				7. Stations: UV/IR flame detection EPOs station and trips fire alarm		E									

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk S	Current Risk P	Current Risk RR	HAZOP Recommendations: Proposed Recommendations	HAZOP Recommendations: Responsibility	Residual Risk S	Residual Risk P	Residual Risk RR	Remarks
				5. Attended operations during fills by trained and qualified operators		T									
				6. Station EPO Button		P									
	3. Operator error - AOV - 104 and NV-104 both left open, passing through line with check valve (CV-301) and no pressure in fill manifold.	1. Potential to misdirecting flow from vent manifold to fill manifold leading to over temperature OR under temperature of tank(s), causing potential for leakage and/or potential ignition of H2	S	13. All tanks have temperatures actively monitored, fault out and close supply valve in the case of over temperature reading. Both the fill valves and de-fueling valves along with the over and under temperature limits.		E	II	D	Low	4. Interlock all tank vent valves with NV-104 being open		II	E	Low	1. The failure mode of the tanks due to adiabatic overheating of the internal hydrogen is to damage the HDPE liner, causing hydrogen permeation (leak-through) of the carbon fiber structure. The tank will NOT rupture under adiabatic heating scenarios, confirmed with manufacturer, and through internally derived calculations and modeling. **The failure mode of the tanks due to adiabatic overheating of the internal hydrogen is to damage the HDPE liner, causing hydrogen permeation (leak-through) of the carbon fiber structure. The tank will NOT rupture under adiabatic heating scenarios, confirmed with manufacturer, and through internally derived calculations and modeling. See zip file including simulations, notes from discussions with**
				14. Due to low thermal mass of H2, unlikely to overshoot tank temperature limits and past alarm set points.		O									
				5. Attended operations during fills by trained and qualified operators		T									
				21. All equipment in the trailer will be classified C1D2		E									
				20. Fixed CG detectors inside the trailer which trigger EMO at 25% LEL - H2		E									

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk			HAZOP Recommendations		Residual Risk			Remarks
							S	P	RR	Proposed Recommendations	Responsibility	S	P	RR	
															manufacturer, and literature papers. Hexagon Purus (tank manufacture) also stated that the tank gas temperature can exceed the stated 85°C limit for no more than 30 hours during the life of the tank. Any temperature excursions above 120°C result in end of life for tank.
		2. Potential to misdirecting flow from vent manifold to fill manifold leading to over temperature OR under temperature of tank(s), causing tanks to operate beyond safe operating limits causing permanent damage. (damage to HDPE liner)	S	13. All tanks have temperatures actively monitored, fault out and close supply valve in the case of over temperature reading. Both the fill valves and de-fueling valves along with the over and under temperature limits.		E	III	D	Low	4. Interlock all tank vent valves with NV-104 being open		III	E	Routine	
				14. Due to low thermal mass of H2, unlikely to overshoot tank temperature limits and past alarm set points.		O									
		3. Potential to misdirect flow from fill manifold to venting tanks leading to over temperature OR under temperature of tank(s), causing tanks to operate beyond safe operating limits causing permanent damage.	S	13. All tanks have temperatures actively monitored, fault out and close supply valve in the case of over temperature reading. Both the fill valves and de-fueling valves along with the over and under temperature limits.		E	II	E	Low						
				14. Due to low thermal mass of H2, unlikely to overshoot tank		O									

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk S	Current Risk P	Current Risk RR	HAZOP Recommendations: Proposed Recommendations	HAZOP Recommendations: Responsibility	Residual Risk S	Residual Risk P	Residual Risk RR	Remarks
				temperature limits and past alarm set points.											
4. Reverse Flow	1. Failure of check valve in the receptacles (H70-HF, H35-HF, H70-LF)	1. Reverse flow from HDHS fueling manifold and tank to dispenser vent leading to significant venting of H2 and high volume of gas venting at the dispenser creating the potential to exceed LEL in immediate area. Creating a potential for H2 ignition and fire.	S	15. Dispenser process vent designed to standard industry practice of restricting flow to avoid depressurizing hose too quickly.		E	I	E	Low						
				16. Dispenser vent designed to CGA 5.5 standard and NFPA 2 (C1D1 around vent termination).		E									
				17. Attended operation by trained and qualified operators.		T									
				18. EMO will return air operated valves to normal position, blocking flow and venting HDHS manifold.		E									
				19. Station fire eye, detecting fire will EPO power supply to HDHS		E									
5. Loss of Containment	1. loose fitting, degraded components due to H2 embrittlement.	1. Release of H2 into trailer, causing accumulation and potential for ignition and fire.	S	20. Fixed CG detectors inside the trailer which trigger EMO at 25% LEL - H2		E	I	D	Moderate	5. Conduct leak check of H2 piping, with inert gas, up to isolation valve (NV-401 through NV-412) after every relocation and prior to testing of the HDHS.		I	E	Low	
				21. All equipment in the trailer will be classified C1D2		E				6. Utilize torque seal and/or indicator to identify if hardware has loosened during transportation.					
				22. Active ventilation with monitored ventilation rate which if fall below set rate -		E									

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk S	Current Risk P	Current Risk RR	HAZOP Recommendations: Proposed Recommendations	HAZOP Recommendations: Responsibility	Residual Risk S	Residual Risk P	Residual Risk RR	Remarks
				will EMO system (minimum of 6 ACH)											
				18. EMO will return air operated valves to normal position, blocking flow and venting HDHS manifold.		E									
				17. Attended operation by trained and qualified operators.		T									
				23. All components of HDHS will be built out of H2 compatible materials and used in accordance with manufacturer specifications.		E									
				24. Trailer: Internal fire eye will trigger fire detection alarm and trips EMO and shuts down ventilation upon detection of H2 fire within HDHS.		E									
	2. Failure of check valve in the receptacles (H70-HF, H35-HF, H70-LF)	1. Release of H2 into ambient, causing accumulation and potential for ignition and fire, worker exposure and/or injury.	S	25. Mechanical interlock does not allow nozzle to release while under pressure, based on nozzle manufacturer design/specs.		E	I			7. After fueling event, decouple the nozzle and then vent down the HDHS manifold prior to recirculating. Close AOV 101-103, limited pressure		I			
				23. All components of HDHS will be built out of H2 compatible materials and used in accordance with manufacturer specifications.		E				8. After fueling event, procedure for operator to initiate (end of fill sequence), reducing H2 pressure and					

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk			HAZOP Recommendations		Residual Risk			Remarks
							S	P	RR	Proposed Recommendations	Responsibility	S	P	RR	
				19. Station fire eye, detecting fire will EPO power supply to HDHS		E				volume by limiting amount of H2 available by blocking up to receptacles. REVISIT TO DETERMINE IF PSV SHOULD BE INCLUDED AND REALITY OF THIS CONTROL.					
6. High Pressure	1. Too high setpoint FCV-103 venting flow control valve, FCV-103 failure.	1. Overpressure of stanchion delivery piping OR vent hose that leads to the vent mast, leading to burst components and release of H2 with potential ignition and fire.	S	26. PT 302 / PSH 302 - monitoring for overpressurization and will trigger EMO and will shut down AOV 101-103 (H2 supply), AOV 201-212 (fueling valves), and AOV 301-312 (de-fueling valves).		E	I	E	Low						
				27. PSV-301 set to the lesser of the hose MAWP or 2,600 psi. PSV will be sized for maximum flow through FCV-103 under worst case conditions (full open with maximum pressure)		E									
	2. Trapped H2 gas at -40C reaching ambient temperature	1. Potential to pressurize line and exceed MAWP of piping and components, leading to burst components and release of H2 with potential ignition and fire.	S	62. All components of HDHS in this node are rated to >/=20000 psig. Highest calculated pressure of trapped with worst case ambient temperature is calculated <20,000 psig (-40C to 40C ambient)		E	I	E	Low	20. Consider adding if TT 101 through 103 see a temperature < -45C the triggers immediate shutdown AOVs 101 thru 103		I	E	Low	
7. Low Pressure	1. Station is not providing enough pressure	1. Operational only, no safety consequence	BI												

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk S	Current Risk P	Current Risk RR	HAZOP Recommendations: Proposed Recommendations	HAZOP Recommendations: Responsibility	Residual Risk S	Residual Risk P	Residual Risk RR	Remarks
8. High Temperature	1. Normal flow from de-fueling.	1. High temperature downstream from FCV-103 caused by J-T heating. Potential to leak H2, leading to ignition and fire.	S	28. Components downstream from FCV-103 are rated 35C above high temperature from the tanks of 85C.		E	I	E	Low						
				7. Stations: UV/IR flame detection EPOs station and trips fire alarm		E									
				20. Fixed CG detectors inside the trailer which trigger EMO at 25% LEL - H2		E									
				21. All equipment in the trailer will be classified C1D2		E									
				18. EMO will return air operated valves to normal position, blocking flow and venting HDHS manifold.		E									
				17. Attended operation by trained and qualified operators.		T									
9. Low Temperature	1. No Credible cause for cryogenic temperatures.														
	2. Station provides H2 < -40C.	1. Low temperature to HDHS components outside of minimum design temperature of components leading to leak of H2, creating potential for ignition and fire.	S	29. If temperature < -40C are reached then system will stop dispensing should close within 5 seconds if normal operating temperatures are not achieved.		E	I	E	Low	20. Consider adding if TT 101 through 103 see a temperature < -45C the triggers immediate shutdown AOVs 101 thru 103		I	E	Low	
				30. If TT 101-103 see a temperature < -40C then AOVs 101-103 will close after 5 seconds.		E									

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk			HAZOP Recommendations		Residual Risk			Remarks
							S	P	RR	Proposed Recommendations	Responsibility	S	P	RR	
				7. Stations: UV/IR flame detection EPOs station and trips fire alarm		E									
				20. Fixed CG detectors inside the trailer which trigger EMO at 25% LEL - H2		E									
				21. All equipment in the trailer will be classified C1D2		E									
				18. EMO will return air operated valves to normal position, blocking flow and venting HDHS manifold.		E									
				17. Attended operation by trained and qualified operators.		T									
10. High Level	1. No level maintained for this node														
11. Low/No Level	1. No level maintained for this node														
12. Other than Composition - Contamination, Impurities, Air	1. During maintenance activities, operator error and does not adequately purge the lines.	1. Creating a flammable mixture within piping, creating a potential for ignition and fire.	S	31. If system of any volume is filled with 100% air and then filled with H2 once the pressure exceeds 44 psig the UEL is exceeded and mixture is no longer flammable.		E	I	E	Low						1. Calculations show that atmospheric entrainment of air, in worse case condition of ignition with H2 at UEL, results in an overpressurization event of 435 psig. Which is well below the MAWP pressure rating of the piping, valves, and instruments.
				32. No ignition sources inside pipes		O									
				33. LOTO and zero-energy verification procedure must be developed for maintenance.		P									

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk			HAZOP Recommendations		Residual Risk			Remarks
							S	P	RR	Proposed Recommendations	Responsibility	S	P	RR	
				34. No pressurized O2 source capable of creating flammable concentrations at pressure		O									
				35. HDHS is bonded and grounded reducing the likelihood of ignition		E									
				36. N2 purge procedures as required by ASME code are required prior to returning system to operation and are detailed in maintenance procedure.		P									
				21. All equipment in the trailer will be classified C1D2		E									
				17. Attended operation by trained and qualified operators.		T									
	2. H2 station quality will be validated by dispensing owner prior to HDHS arrival. No credible cause.														
13. Other than Sequence - Wrong action in operation, Operation not completed	1. Operator error - incorrect installation of vent mast OR forgetting to connect vent mast	1. Venting high pressure H2 near operators creating the potential for high pressure injury and creating potential for ignition and fire.	S	37. SOP includes checklist verification of all ancillary equipment connections		PPE	I	D	Moderate	11. Conduct leak check of H2 piping, with inert gas, up to isolation valve (NV-401 through NV-412) after every relocation and prior to testing of the HDHS. This will be the first indication if the vent mast is not installed properly, leaking only inert gas.		I	E	Low	

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk S	Current Risk P	Current Risk RR	HAZOP Recommendations: Proposed Recommendations	HAZOP Recommendations: Responsibility	Residual Risk S	Residual Risk P	Residual Risk RR	Remarks
				17. Attended operation by trained and qualified operators.		T				9. Relocate NV-104 downstream from de-fueling vent prior to FCV-103 to use an administrative lock that will be placed during shutdown.					
				38. PPE including FR clothing, safety glasses,		PPE				10. Include an electrical interlock which will close a loop only once the vent mast is installed (manual connection).					
										2. Recommend operators PPE to include personal CG/O2 detector and FR clothing and FR gloves when dispensing or conducting maintenance of HDHS.					
14. Control System Failure	1. Safety critical sensors or instruments fail or malfunction	1. Loss of a safety critical shutdown creating the potential to leaking of H2, potential for ignition and fire.	S	39. All safety critical sensors will lead to fail safe condition upon failure or malfunction (e.g., out of range alarm, IO module watchdog), tripping EMO and shuts down.		E	I	E	Low						
				17. Attended operation by trained and qualified operators.		T									
	2. Control system program/software failure or malfunction, hung program.	1. valves would not be responding to commands from control system creating a loss of communication.	S	40. EMO, CGs, and flame detection relays are hardwired to remove control power to control valves, operating independent of PLC.		E	II	D	Low	12. Use a safety rated PLC control OR use a redundant watchdog system (if one system goes down or loses communication the		II	E	Low	

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk			HAZOP Recommendations		Residual Risk			Remarks
							S	P	RR	Proposed Recommendations	Responsibility	S	P	RR	
				17. Attended operation by trained and qualified operators.		T				whole process shuts down).					
15. Power Failure	1. Loss of power to control FCV-103 and FCV-101	1. Inability to control flow to tanks and/or vent, operational only.	BI												
	2. EPO of station shutting down power to HDHS	1. EPO shuts down power to HDHS leading to loss of ventilation and accumulation of H2 creating potential ignition sources and fire.	S	43. All control system power is removed leading to valves close to safe state configuration.		E	I	E	Low	13. SOP will include do not enter HDHS upon power failure and loss of monitoring equipment.		I	E	Low	
				44. Tank pressure gauges will be visible externally from the HDHS.		E				17. SOP no worker should open and/or enter HDHS until safe atmosphere is safely confirmed.					
				45. Select fans (based on H2 gas accumulation) will be C1D1 rated.		E									
				47. Without power, less likely to have an ignition source in the HDHS.		O									
				35. HDHS is bonded and grounded reducing the likelihood of ignition		E									
				48. C1D1 rated CG relay, UV/IR fire detection AND loss of ventilation relay will not allow non C1D1 equipment to power up until a safe atmosphere is verified.		E									
				49. Control screen and beacon will remain off indicating loss of power to the system.		E									

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk			HAZOP Recommendations		Residual Risk			Remarks
							S	P	RR	Proposed Recommendations	Responsibility	S	P	RR	
				50. Process Beacon/light tree will be installed on the exterior of the HDHS to indicate process conditions (e.g., running, EPO/EMO)		E									
				51. C1D1 CG beacon/light tree will be installed on the exterior of the HDHS to indicate status of CG detection (combustible gas detection indication at/above high-high condition)		E									
		2. Loss of control and monitoring equipment (e.g., fire eye, CG detector) with potential for unknown H2 accumulation. Accumulation of H2 creating potential ignition sources and fire.	S	52. Same as above, "EPO shutdown..." Safeguards.		O	I	E	Low	15. SOP no worker should open and/or enter HDHS until safe atmosphere is safely confirmed.		I	E	Low	
				53. CG and ventilation will be hardwired to autostart upon return of power.		E				16. Ensure ventilation equipment is C1D1 rated					
										14. Upon reenergization of HDHS, if C1D1 rated CG relays identified unsafe atmosphere exterior indicator signal will be activated to alert workers to hazardous atmosphere.					
16. Operator Interaction	1. Connecting dispenser to HDHS	1. Creating a potential leak for H2 creating potential ignition source and fire.	S	17. Attended operation by trained and qualified operators.		T	II	D	Low	18. Workers should wear personal CG monitors		II	E	Low	
				54. HDHS operators will be wearing FR clothing		PPE									

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk			HAZOP Recommendations		Residual Risk			Remarks
							S	P	RR	Proposed Recommendations	Responsibility	S	P	RR	
				55. Dispenser station internal cabinet is equipped with CG detection per code requirements		E									
				7. Stations: UV/IR flame detection EPOs station and trips fire alarm		E									

Node: 2. H2 Storage Tanks

Drawings / References:

Design Conditions/Parameters: MAOP: xx MPa, MAWP: xx MPa;
MAOT: -40C min, 85C max;
Flow: 20 kg/min peak, 10 kg/min avg
Hydrogen

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk			HAZOP Recommendations		Residual Risk			Remarks
							S	P	RR	Proposed Recommendations	Responsibility	S	P	RR	
1. High Flow	1. Dispenser providing flow outside of pressure ramp rate (>20 kg/min)	1. Overtemperature of tanks with potential to exceed MAWT of H2 storage tank liners, damaging tanks due to operation outside of design envelope. Potential for H2 leak and fire.	S	1. Fueling protocol dictates pressure ramp rates and if falls outside window above the pressure ramp rate high-high, then the station should shut down the fill within 5 seconds.		E	I	E	Low						1.
				2. Dispenser meets CSA HGV-4.3 H2 fueling parameters verification of station		E									
				56. TT-401 A/B through TT-412 A/B, High-high temperature alarms triggering a process stop. (P-stop)		E									
				22. Active ventilation with monitored ventilation rate which if fall below set rate - will EMO system (minimum of 6 ACH)		E									
				20. Fixed CG detectors inside the trailer which trigger EMO at 25% LEL - H2		E									
				57. TT-401 A/B through TT-412 A/B, High temperature alarm will notify trained operators.		E									
				5. Attended operations during fills by trained and qualified operators		T									

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk S	Current Risk P	Current Risk RR	HAZOP Recommendations: Proposed Recommendations	HAZOP Recommendations: Responsibility	Residual Risk S	Residual Risk P	Residual Risk RR	Remarks
				58. The failure mode of the tanks due to adiabatic overheating of the internal hydrogen is to damage the HDPE liner, causing hydrogen permeation (leak-through) of the carbon fiber structure. The tank will NOT rupture under adiabatic heating scenarios, confirmed with manufacturer, and through internally derived calculations and modeling. See zip file including simulations, notes from discussions with manufacturer, and literature papers.		E									
				24. Trailer: Internal fire eye will trigger fire detection alarm and trips EMO and shuts down ventilation upon detection of H2 fire within HDHS.		E									
	2. Operator error - Defueling tanks too quickly	1. Adiabatic cooling with potential to exceed MDMT damage H2 storage tanks, leading to potential for future leaking out of tank composite liner, H2 ignition and fire	S	63. FCV-103 flow control valve automatically controls based on feedback pressure decay rates of H2 Storage PT-401A through 412A that throttles venting		E	II	E	Low						
				64. Allowable blowdown rate exceeded (calculated from pressure decay PT-401A through 412A and are tank specific), automatically shuts down AOV-401 thru 412		E									
				22. Active ventilation with monitored ventilation rate which if fall below set rate -		E									

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk S	Current Risk P	Current Risk RR	HAZOP Recommendations: Proposed Recommendations	HAZOP Recommendations: Responsibility	Residual Risk S	Residual Risk P	Residual Risk RR	Remarks
				will EMO system (minimum of 6 ACH)											
				20. Fixed CG detectors inside the trailer which trigger EMO at 25% LEL - H2		E									
				5. Attended operations during fills by trained and qualified operators		T									
				24. Trailer: Internal fire eye will trigger fire detection alarm and trips EMO and shuts down ventilation upon detection of H2 fire within HDHS.		E									
				58. The failure mode of the tanks due to adiabatic overheating of the internal hydrogen is to damage the HDPE liner, causing hydrogen permeation (leak-through) of the carbon fiber structure. The tank will NOT rupture under adiabatic heating scenarios, confirmed with manufacturer, and through internally derived calculations and modeling. See zip file including simulations, notes from discussions with manufacturer, and literature papers.		E									
2. No/Low Flow	1. Industry is reviewing effects of thermal stratification within the tanks. NREL research has identified low ramp														

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk			HAZOP Recommendations		Residual Risk			Remarks
							S	P	RR	Proposed Recommendations	Responsibility	S	P	RR	
	rate leading to potential thermal stratification and elevated temperatures above bulk gas limit. No further safety consequence.														
3. Misdirected Flow	1. Operator error, wrong set of valves being opened.	1. Potential to misdirecting flow from a full tank to an empty tank, leading to over temperature OR under temperature of tank(s), causing tanks to operate beyond safe operating limits causing permanent damage. (damage to HDPE liner)	S	13. All tanks have temperatures actively monitored, fault out and close supply valve in the case of over temperature reading. Both the fill valves and de-fueling valves along with the over and under temperature limits.		E	III	E	Routine						
				14. Due to low thermal mass of H2, unlikely to overshoot tank temperature limits and past alarm set points.		O									
				56. TT-401 A/B through TT-412 A/B, High-high temperature alarms triggering a process stop. (P-stop)		E									
				57. TT-401 A/B through TT-412 A/B, High temperature alarm will notify trained operators.		E									
				59. TT-401 A/B through TT-412 A/B, Low-low temperature alarms triggering a process stop. (P-stop)		E									
				60. TT-401 A/B through TT-412 A/B, Low temperature alarm will notify trained operators.		E									

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk			HAZOP Recommendations		Residual Risk			Remarks
							S	P	RR	Proposed Recommendations	Responsibility	S	P	RR	
		2. Send to H2 to wrong sized tank, operational only, no safety consequence.	BI												
4. Reverse Flow	1. No additional cause.														
5. Loss of Containment	1. loose fitting, degraded components due to H2 embrittlement.	1. Release of H2 into trailer, causing accumulation and potential for ignition and fire.	S	24. Trailer: Internal fire eye will trigger fire detection alarm and trips EMO and shuts down ventilation upon detection of H2 fire within HDHS.		E	I	D	Moderate	5. Conduct leak check of H2 piping, with inert gas, up to isolation valve (NV-401 through NV-412) after every relocation and prior to testing of the HDHS.		I	E	Low	
				20. Fixed CG detectors inside the trailer which trigger EMO at 25% LEL - H2		E				6. Utilize torque seal and/or indicator to identify if hardware has loosened during transportation.					
				21. All equipment in the trailer will be classified C1D2		E									
				22. Active ventilation with monitored ventilation rate which if fall below set rate - will EMO system (minimum of 6 ACH)		E									
				18. EMO will return air operated valves to normal position, blocking flow and venting HDHS manifold.		E									
				17. Attended operation by trained and qualified operators.		T									
				23. All components of HDHS will be built out of H2 compatible materials and used in		E									

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk			HAZOP Recommendations		Residual Risk			Remarks
							S	P	RR	Proposed Recommendations	Responsibility	S	P	RR	
				accordance with manufacturer specifications.											
	2. Operator Error, NV-400 (drain valve) left open or uncapped.	1. Release of TPRD vent, releasing H2 into the trailer.	S	61. TPRD is a mechanical device that is only activated when TPRD exceeds temperature setpoint - manufacturers set point set by OEM		E	I	D	Moderate	19. TPRD vent stack draining only occurs when HDHS is below 30 psig. Maintenance procedures should not allow draining vent mast at tank pressures above the travel pressure of 30 psig and verifying the draining cap has been installed correctly.		I	E	Low	1.
				17. Attended operation by trained and qualified operators.		T									
				20. Fixed CG detectors inside the trailer which trigger EMO at 25% LEL - H2		E									
				21. All equipment in the trailer will be classified C1D2		E									
				22. Active ventilation with monitored ventilation rate which if fall below set rate - will EMO system (minimum of 6 ACH)		E									
				18. EMO will return air operated valves to normal position, blocking flow and venting HDHS manifold.		E									
				23. All components of HDHS will be built out of H2 compatible materials and used in accordance with manufacturer specifications.		E									
6. High Pressure	1. Station provides pressure exceeding MAWP of 96.25 MPa	1. Exceed pressure rating of H2 Storage Tank, leading to rupture,	S	70. Calculation of H2 gas density used to calculate tank specific SOC. SOC is designed to limit the end of fill density of H2 in		E	I	E	Low						

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk			HAZOP Recommendations		Residual Risk			Remarks
							S	P	RR	Proposed Recommendations	Responsibility	S	P	RR	
		potential for leak, H2 ignition and fire		tanks. Calculation is based on most conservative of PT-401A through PT-412A AND of TT-401A/B through TT-412A/B upon High-High SOC for the tank, P-Stop of the station											
				71. Station Dispenser H70 fueling MOP 87.5 MPa (12690 psig) and Station H70 fueling PSVs set at maximum of 96.25 MPa (13960 psig)		E									
				72. All non-tank components rated to a minimum of 20000 psig		E									
				73. H2 Storage Tanks are DOT rated and the minimum Test Pressure is rated to least the highest MAWP that can be set for the Dispenser Station of 96.25 MPa (13960 psig)		E									
	2. Cold pressurized gas coasts up to ambient temperatures - Not a credible scenario in this node														
7. Low Pressure	1. No safety consequence of interest with low pressure in this node														
8. High Temperature	1. Engulfing fire or adjacent fire	1. Potential to exceed MAWP of H2 Storage Tanks, leading to loss of tank structural integrity and catastrophic loss of containment. H2 ignition, fire, explosion	S	7. Stations: UV/IR flame detection EPOs station and trips fire alarm		E	I	E	Low						
				67. Fueling station designed to NFPA 2 Gaseous Fueling Facilities and IFC		E									

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk			HAZOP Recommendations		Residual Risk			Remarks
							S	P	RR	Proposed Recommendations	Responsibility	S	P	RR	
				68. Separation distance of 30' from wildland areas per NFPA 1140 Standard for Wildland Fire Protection		E									
				65. All TTs in the HDHS system have high-high temperature alarms and fault alarms triggering a process stop. (P-stop)		E									
				66. Trailer interior to be clad in non-flammable materials (e.g., metal sheeting). Limited combustible material.		E									
				61. TPRD is a mechanical device that is only activated when TPRD exceeds temperature setpoint - manufacturers set point set by OEM		E									
				24. Trailer: Internal fire eye will trigger fire detection alarm and trips EMO and shuts down ventilation upon detection of H2 fire within HDHS.		E									
				17. Attended operation by trained and qualified operators.		T									
				21. All equipment in the trailer will be classified C1D2		E									
	2. Dispenser H2 feed is too high temperature (ambient temperature)	1. H2 flow from adiabatic heating leading to exceed MAWT of H2 Storage Tanks, leading to Overtemperature of tanks with potential to exceed MAWT of H2	S	69. Fueling protocol dictates flow rate based on inlet H2 temperature		E	I	E	Low	21. Consider early shutdown of fueling valve AOV-101, 102, 103 based on H2 high temperature from station as prescribed					

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk			HAZOP Recommendations		Residual Risk			Remarks
							S	P	RR	Proposed Recommendations	Responsibility	S	P	RR	
		storage tank liners, damaging tanks due to operation outside of design envelope. Potential for H2 leak and fire. **Additional Safeguards listed in High Flow Above – Dispenser providing flow outside of pressure ramp rate (>20 kg/min)**								in developing Fueling Protocol					
9. Low Temperature	1. No further cause **See Node 1 Low Temperature and Node 2 High Defueling Flow**														
10. High Level	1. See High Pressure for High SOC considerations														
11. Low/No Level	1. No consequence of interest														
12. Other than Composition - Contamination, Impurities, Air	1. Same as Node 1														
13. Other than Sequence - Wrong action in operation, Operation not completed	1. Incorrect system configuration sent to dispenser for fueling	1. Incorrect tanks in fill with different volume and ramp rate, potential for prescribed incorrect ramp rate leading to overtemperature of tank, exceeding MAWT, potential for H2 leak, fire, ignition	S	74. High-high and low-low alarms are always active for tanks, even if not intended for use		E									

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk			HAZOP Recommendations		Residual Risk			Remarks
							S	P	RR	Proposed Recommendations	Responsibility	S	P	RR	
		Additional Safeguards listed in High Flow Above – Dispenser providing flow outside of pressure ramp rate (>20 kg/min)													
14. Control System Failure	1. Operator error - sensor disabled	1. Loss of a safety critical shutdown creating the potential for leaking of H2, potential for ignition and fire.	S	75. Out of Range fault activated when sensor unplugged		E									
				76. Password protecting control software to ensure safety system integrity according to cause and effects matrix		E									
	2. Same as Node 1														
15. Power Failure	1. Same as Node 1														
16. Operator Interaction	1. Operator has no routine operating interaction in this Node														

Node: 3. Inert Gas System

Drawings / References:

Design Conditions/Parameters: Purge N2: MAOP/MAWP: cylinder pressure 2500 psig; MAOT ambient temperatures
Instrumentation N2: MAOP/MAWP: 125 psig; MAOT: ambient temperatures

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk S	Current Risk P	Current Risk RR	HAZOP Recommendations: Proposed Recommendations	HAZOP Recommendations: Responsibility	Residual Risk S	Residual Risk P	Residual Risk RR	Remarks
1. High Flow	1. No safety consequence of high flow, see high pressure.	1.	BI												
2. No/Low Flow	1. Loss of N2 or power to control FCV-103, FCV-101	1. Inability to open or close any control valves, some may be partially open OR closed when thought to be open, leading to uncontrolled pressure equalization. Additionally creating the potential for misdirecting flow from a full tank to an empty tank, leading to over temperature OR under temperature of tank(s), causing tanks to operate beyond safe operating limits causing permanent damage. (damage to HDPE liner)	S	77. Ensure pathway to vent for fuel and de-fuel manifolds under power loss or loss of N2.		E	III	E	Routine	22. Consider adding P-stop to PSL-702					
				41. All air operated shutoff valves return to safe state upon the loss of pressure from N2.		E				23. Determine if it is necessary to vent the fuel and de-fuel manifolds under certain safety conditions. Will determine how to proceed with existing safeguards 84.					
				42. PSL-702 low pressure N2 switch will trigger an alarm if low/no pressure.		E									
				17. Attended operation by trained and qualified operators.		T									
				13. All tanks have temperatures actively monitored, fault out and close supply valve in the case of over temperature reading. Both the fill valves and de-fueling valves along with the over and under temperature limits.		E									
				14. Due to low thermal mass of H2, unlikely to overshoot tank temperature limits and past alarm set points.		O									
				56. TT-401 A/B through TT-412 A/B, High-high temperature		E									

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk			HAZOP Recommendations		Residual Risk			Remarks
							S	P	RR	Proposed Recommendations	Responsibility	S	P	RR	
				alarms triggering a process stop. (P-stop)											
				57. TT-401 A/B through TT-412 A/B, High temperature alarm will notify trained operators.		E									
				59. TT-401 A/B through TT-412 A/B, Low-low temperature alarms triggering a process stop. (P-stop)		E									
				60. TT-401 A/B through TT-412 A/B, Low temperature alarm will notify trained operators.		E									
				86. Startup procedure will include N2 cylinder volume check		P									
3. Misdirected Flow	1. Operator error - Connecting pneumatic lines improperly to wrong AOV valve. Misalignment of fueling protocol to vehicle tank volume	1. Misalignment of fueling protocol to vehicle tank volume leading to potential overtemp causing tanks to operate beyond safe operating limits causing permanent damage. (damage to HDPE liner)	S	78. Trained and qualified workers performing maintenance operations.		T	III	E	Routine						
				13. All tanks have temperatures actively monitored, fault out and close supply valve in the case of over temperature reading. Both the fill valves and de-fueling valves along with the over and under temperature limits.		E									
				14. Due to low thermal mass of H2, unlikely to overshoot tank temperature limits and past alarm set points.		O									
				56. TT-401 A/B through TT-412 A/B, High-high temperature		E									

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk			HAZOP Recommendations		Residual Risk			Remarks
							S	P	RR	Proposed Recommendations	Responsibility	S	P	RR	
				alarms triggering a process stop. (P-stop)											
				57. TT-401 A/B through TT-412 A/B, High temperature alarm will notify trained operators.		E									
				59. TT-401 A/B through TT-412 A/B, Low-low temperature alarms triggering a process stop. (P-stop)		E									
				60. TT-401 A/B through TT-412 A/B, Low temperature alarm will notify trained operators.		E									
4. Reverse Flow	1. Operator error - NV-601, NV-602, and check valve CV-606 are left in open position.	1. Potential to send H2 to N2 lines leading to exceeding MAWP, burst components, release of H2, potential ignition of H2/fire inside HDHS.	S	79. PSV-601 set to 2890 psig and venting to a safe location.		E	I	E	Low						
				80. Rupture disc (BD-601) set at 3500 psi and venting to a safe location. Sized for chocked flow through 601 and 602 at MOP.		E									
				17. Attended operation by trained and qualified operators.		T									
				81. PSH-601 high pressure (set higher than N2 cylinder) will EMO HDHS.		E									
	2. H2 to air operated valve pneumatic line.	1. Potential to send H2 to N2 lines leading to exceeding MAWP, burst components, release of H2, potential ignition of H2/fire inside HDHS.	S	82. Ensure sufficient air gap between process control and pneumatic operation in selection of H2 AOVs.		E	I	E	Low	24. Industry standard H2 process control valves include designs similar to autoclave engineer SM model valves.					
				24. Trailer: Internal fire eye will trigger fire detection alarm and trips EMO and shuts down		E									

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk			HAZOP Recommendations		Residual Risk			Remarks
							S	P	RR	Proposed Recommendations	Responsibility	S	P	RR	
				ventilation upon detection of H2 fire within HDHS.											
				20. Fixed CG detectors inside the trailer which trigger EMO at 25% LEL - H2		E									
				21. All equipment in the trailer will be classified C1D2		E									
				22. Active ventilation with monitored ventilation rate which if fall below set rate - will EMO system (minimum of 6 ACH)		E									
				18. EMO will return air operated valves to normal position, blocking flow and venting HDHS manifold.		E									
				17. Attended operation by trained and qualified operators.		T									
5. Loss of Containment	1. loose fittings	1. Loss of N2, simple asphyxiant inside trailer and 02 deficiency inside HDHS.	S	22. Active ventilation with monitored ventilation rate which if fall below set rate - will EMO system (minimum of 6 ACH)		E	I	E	Low	2. Recommend operators PPE to include personal CG/O2 detector and FR clothing and FR gloves when dispensing or conducting maintenance of HDHS.					
				83. N2 limited to volume of one cylinder (approximately 256 cu ft)		O									
				84. Trailer is not intended for human occupancy during operating conditions.		P									
6. High Pressure	1. Pressure regular failure PRV-701	1. Overpressure instrument nitrogen to	S	85. PSV-701 set at 125 psig, sized for the full flow of the		E	I	E	Low	2. Recommend operators PPE to include personal					

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk			HAZOP Recommendations		Residual Risk			Remarks
							S	P	RR	Proposed Recommendations	Responsibility	S	P	RR	
		AOV valves, potential to exceed MAWP of the instrument nitrogen lines and/or air operators. Leading to potential burst components and N2 to trailer (O2 deficiency inside HDHS)		regulator. Relief vented routed externally to safe location						CG/O2 detector and FR clothing and FR gloves when dispensing or conducting maintenance of HDHS.					
				82. Ensure sufficient air gap between process control and pneumatic operation in selection of H2 AOVs.		E									
				22. Active ventilation with monitored ventilation rate which if fall below set rate - will EMO system (minimum of 6 ACH)		E									
				83. N2 limited to volume of one cylinder (approximately 256 cu ft)		O									
				84. Trailer is not intended for human occupancy during operating conditions.		P									
	2. See Reverse Flow for "Bleed back of H2 into N2 line"														
7. Low/Vacuum Pressure	1. Deplete cylinder	1. Inability to open or close any control valves, some may be partially open OR closed when thought to be open, leading to uncontrolled pressure equalization. Additionally creating the potential for misdirecting flow from a full tank to an empty tank, leading to over temperature OR under temperature of tank(s),	S												

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk			HAZOP Recommendations		Residual Risk			Remarks
							S	P	RR	Proposed Recommendations	Responsibility	S	P	RR	
		causing tanks to operate beyond safe operating limits causing permanent damage. (damage to HDPE liner) **See Low/No Flow Above**													
		2. Inability to inert as required. Operational only, will require delayed startup	BI												
8. High Temperature	1. No credible scenario for high temperature														
9. Low/Cryogenic Temperature	1. Inerting quickly into interconnection piping or flow quickly into nitrogen air-operated lines	1. High flow, with cold inert gas into H2 or N2 line from 2500 psi cylinder, temperature fall below MAWT of line, Potential for leak of N2 into HDHS. Potential for displace of air, operator exposure O2 deficiency air	S	87. Lines 6001 to 6006 and components rated to -40C		E	I	D	Moderate	2. Recommend operators PPE to include personal CG/O2 detector and FR clothing and FR gloves when dispensing or conducting maintenance of HDHS.		I	E	Low	
				22. Active ventilation with monitored ventilation rate which if fall below set rate - will EMO system (minimum of 6 ACH)		E									
				84. Trailer is not intended for human occupancy during operating conditions.		P									
				83. N2 limited to volume of one cylinder (approximately 256 cu ft)		O									
				88. Operating procedures require the doors to be left open when anyone is inside		P									
10. High Level	1. No level maintained in this node														

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk S	Current Risk P	Current Risk RR	HAZOP Recommendations: Proposed Recommendations	HAZOP Recommendations: Responsibility	Residual Risk S	Residual Risk P	Residual Risk RR	Remarks
11. Low/No Level	1. No level maintained in this node														
12. Other than Composition - Contamination, Impurities, Air	1. Reverse flow of H2 in N2 lines	1. Contamination of N2 cylinder and N2 lines with H2. When purging - possible mixture of N2/H2 in line leading to inadequate inerting with personal exposure to ambient pressure H2 during maintenance. Potential for ignition, fire.	S	89. Procedure requires PI-601 must read 0 psig before needle valves (NV-601 and NV-602) from N2 cylinder are opened		P	II	D	Low	25. Develop response procedure for contaminated N2 cylinder system.		II	E	Low	
				33. LOTO and zero-energy verification procedure must be developed for maintenance.		P									
				78. Trained and qualified workers performing maintenance operations.		T									
				90. If PSH-601 or PSV-601 are activated, operator must vent H2 system and conduct root cause analysis and determine cause of failure and mitigate before operation may continue.		P									
				81. PSH-601 high pressure (set higher than N2 cylinder) will EMO HDHS.		E									
				79. PSV-601 set to 2890 psig and venting to a safe location.		E									
		2. Contamination of N2 cylinder and N2 lines with H2. When air-operated valves are actuated, possible mixture of N2/H2 into HDHS. Potential for H2	S	89. Procedure requires PI-601 must read 0 psig before needle valves (NV-601 and NV-602) from N2 cylinder are opened		P	I	E	Low	25. Develop response procedure for contaminated N2 cylinder system.		I	E	Low	
				90. If PSH-601 or PSV-601 are activated, operator must vent		P									

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk			HAZOP Recommendations		Residual Risk			Remarks
							S	P	RR	Proposed Recommendations	Responsibility	S	P	RR	
		accumulation, ignition and fire		H2 system and conduct root cause analysis and determine cause of failure and mitigate before operation may continue.											
				21. All equipment in the trailer will be classified C1D2		E									
				22. Active ventilation with monitored ventilation rate which if fall below set rate - will EMO system (minimum of 6 ACH)		E									
				20. Fixed CG detectors inside the trailer which trigger EMO at 25% LEL - H2		E									
				40. EMO, CGs, and flame detection relays are hardwired to remove control power to control valves, operating independent of PLC.		E									
				51. C1D1 CG beacon/light tree will be installed on the exterior of the HDHS to indicate status of CG detection (combustible gas detection indication at/above high-high condition)		E									
				48. C1D1 rated CG relay, UV/IR fire detection AND loss of ventilation relay will not allow non C1D1 equipment to power up until a safe atmosphere is verified.		E									

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk			HAZOP Recommendations		Residual Risk			Remarks
							S	P	RR	Proposed Recommendations	Responsibility	S	P	RR	
				53. CG and ventilation will be hardwired to autostart upon return of power.		E									
	2. Incorrect K bottle connection - CGA fitting prevent flammable or oxidizer being connected	1. Different inert i.e. helium or mixture of inert gases connected. Operational only, no safety consequence	BI												
13. Other than Sequence - Wrong action in operation, Operation not completed	1. Inadequate purging	1. Potential for hydrogen remaining in line. **See Other than Composition Above - Question 12, Consequence 1**	S												
14. Control System Failure	1. Safety critical sensors or instruments fail or malfunction (PSH-601 OR PSL-702)	1. Loss of a safety critical shutdown creating the potential to leaking of H2, potential for ignition and fire.	S	39. All safety critical sensors will lead to fail safe condition upon failure or malfunction (e.g., out of range alarm, IO module watchdog), tripping EMO and shuts down.		E	I	E	Low						
				17. Attended operation by trained and qualified operators.		T									
				79. PSV-601 set to 2890 psig and venting to a safe location.		E									
				80. Rupture disc (BD-601) set at 3500 psi and venting to a safe location. Sized for chocked flow through 601 and 602 at MOP.		E									
				41. All air operated shutoff valves return to safe state upon the loss of pressure from N2.		E									

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk			HAZOP Recommendations		Residual Risk			Remarks
							S	P	RR	Proposed Recommendations	Responsibility	S	P	RR	
15. Power Failure	1. Within this node, no additional safety consequences														
16. Operator Interaction	1. Switching out cylinders - Design of ramp, ergonomics is outside of scope of this PHA														
	2. Inerting before Maintenance - Lockout/Tagout Trained Operators with Equipment Specific LOTO procedures is outside of scope of this PHA														
	3. Leak testing -- Operating procedures for Leak Testing to be developed and outside scope of this PHA														

Node: 4. Trailer

Drawings / References:

Design Conditions/Parameters:

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk			HAZOP Recommendations		Residual Risk			Remarks
							S	P	RR	Proposed Recommendations	Responsibility	S	P	RR	
1. Transportation - Vehicle Incident	1. Vehicle/HDHS crash	1. Potential for hydrogen tubing, storage tank damage with leak of H2, potential for fire, ignition - **This will not be risk rated as out of scope - part of Trailer Design**	S	91. HDHS blown down less than 30 psig, remaining H2 in tanks up to 600 g of H2 (250 scf) in HDHS		P									
				92. Tanks must be mounted according to manufacturer's recommendation		E									
				93. Mounting of tubing and components to DOT codes and recommendations		E									
				94. Fan louvers provide passive ventilation		O									
2. External Fire	1. See H2 Storage Tanks Node, Deviation High Temperature, Engulfing Fire or External Fire														
3. Operator Interaction	1. Set up Process and Emergency Vents - consider operator in design														
	2. Leak Test Before Operation - develop leak test procedure for startup														
	3. Start-up Checklist to include N2 cylinder status, to include Station nozzle hose connection - develop startup of station checklist														

Deviation	Potential Causes	Potential Consequences	CCAT	Existing Safeguards	SCC	SCAT	Current Risk			HAZOP Recommendations		Residual Risk			Remarks
							S	P	RR	Proposed Recommendations	Responsibility	S	P	RR	
	4. Shutdown Operating Checklist - venting down storage to 30 psig - develop shutdown of operation checklist														
4. Public Interaction	1. Emergency response for station inadequate or not in place - develop communication checklist or requirements for HDHS to be shared with First Responders														

HAZOP Safeguards (Master List)

Safeguards	Safety Critical Control	Category	Maximum Current Severity	Place(s) Used
1. Fueling protocol dictates pressure ramp rates and if falls outside window above the pressure ramp rate high-high, then the station should shut down the fill within 5 seconds.		E	I	Consequences: 1.3.1.1, 1.3.1.2, 2.1.1.1
2. Dispenser meets CSA HGV-4.3 H2 fueling parameters verification of station		E	I	Consequences: 2.1.1.1
3. PSV-101 H35 relief valve set @ 6500 psig (45 MPa) sized for flow of H70 HF fill, vented to CGA G-5.5 emergency vent stack		E	I	Consequences: 1.3.1.1

Safeguards	Safety Critical Control	Category	Maximum Current Severity	Place(s) Used
4. PSH-102/PT-102 high pressure initiates emergency stop fill (EMO) to HD HyStEP. Action AOV-201 through 212 close and initiates pressure to spike outside of the pressure ramp window.		E	I	Consequences: 1.3.1.1, 1.3.1.2
5. Attended operations during fills by trained and qualified operators		T	I	Consequences: 1.3.1.1, 1.3.1.2, 1.3.2.2, 1.3.3.1, 2.1.1.1, 2.1.2.1
6. Station EPO Button		P	I	Consequences: 1.3.1.1, 1.3.1.2, 1.3.2.2
7. Stations: UV/IR flame detection EPOs station and trips fire alarm		E	I	Consequences: 1.3.1.1, 1.3.1.2, 1.3.2.2, 1.8.1.1, 1.9.2.1, 1.16.1.1, 2.8.1.1
8. Hazard Classified area Class 1, Div 2 (15 feet for dispenser) with immediate vent path Class 1, Div 1 (5 feet)		E	I	Consequences: 1.3.1.2, 1.3.2.2
9. Station dispenser fixed gas detector for LEL-H2 at 50% of LEL initiates Station EPO		E	I	Consequences: 1.3.1.2, 1.3.2.2
10. Emergency PSV vent designed to CGA G-5.5		E	I	Consequences: 1.3.1.2
11. NFPA 2 setback for emergency ventilation met		E	I	Consequences: 1.3.1.2
12. Receptacle cycle life counter implemented (PT-101, PT-102 and PT-103) and receptacle changed out when cycle life exceeded.		P	I	Consequences: 1.3.2.2
13. All tanks have temperatures actively monitored, fault out and close supply valve in the		E	II	Consequences: 1.3.3.1, 1.3.3.2, 1.3.3.3, 2.3.1.1, 3.2.1.1, 3.3.1.1

Safeguards	Safety Critical Control	Category	Maximum Current Severity	Place(s) Used
case of over temperature reading. Both the fill valves and de-fueling valves along with the over and under temperature limits.				
14. Due to low thermal mass of H2, unlikely to overshoot tank temperature limits and past alarm set points.		O	II	Consequences: 1.3.3.1, 1.3.3.2, 1.3.3.3, 2.3.1.1, 3.2.1.1, 3.3.1.1
15. Dispenser process vent designed to standard industry practice of restricting flow to avoid depressurizing hose too quickly.		E	I	Consequences: 1.4.1.1
16. Dispenser vent designed to CGA 5.5 standard and NFPA 2 (C1D1 around vent termination).		E	I	Consequences: 1.4.1.1
17. Attended operation by trained and qualified operators.		T	I	Consequences: 1.4.1.1, 1.5.1.1, 1.8.1.1, 1.9.2.1, 1.12.1.1, 1.13.1.1, 1.14.1.1, 1.14.2.1, 1.16.1.1, 2.5.1.1, 2.5.2.1, 2.8.1.1, 3.2.1.1, 3.4.1.1, 3.4.2.1, 3.14.1.1
18. EMO will return air operated valves to normal position, blocking flow and venting HDHS manifold.		E	I	Consequences: 1.4.1.1, 1.5.1.1, 1.8.1.1, 1.9.2.1, 2.5.1.1, 2.5.2.1, 3.4.2.1
19. Station fire eye, detecting fire will EPO power supply to HDHS		E	I	Consequences: 1.4.1.1, 1.5.2.1
20. Fixed CG detectors inside the trailer which trigger EMO at 25% LEL - H2		E	I	Consequences: 1.3.3.1, 1.5.1.1, 1.8.1.1, 1.9.2.1, 2.1.1.1, 2.1.2.1, 2.5.1.1, 2.5.2.1, 3.4.2.1, 3.12.1.2
21. All equipment in the trailer will be classified C1D2		E	I	Consequences: 1.3.3.1, 1.5.1.1, 1.8.1.1, 1.9.2.1, 1.12.1.1, 2.5.1.1, 2.5.2.1, 2.8.1.1, 3.4.2.1, 3.12.1.2

Safeguards	Safety Critical Control	Category	Maximum Current Severity	Place(s) Used
22. Active ventilation with monitored ventilation rate which if fall below set rate - will EMO system (minimum of 6 ACH)		E	I	Consequences: 1.5.1.1, 2.1.1.1, 2.1.2.1, 2.5.1.1, 2.5.2.1, 3.4.2.1, 3.5.1.1, 3.6.1.1, 3.9.1.1, 3.12.1.2
23. All components of HDHS will be built out of H2 compatible materials and used in accordance with manufacturer specifications.		E	I	Consequences: 1.5.1.1, 1.5.2.1, 2.5.1.1, 2.5.2.1
24. Trailer: Internal fire eye will trigger fire detection alarm and trips EMO and shuts down ventilation upon detection of H2 fire within HDHS.		E	I	Consequences: 1.5.1.1, 2.1.1.1, 2.1.2.1, 2.5.1.1, 2.8.1.1, 3.4.2.1
25. Mechanical interlock does not allow nozzle to release while under pressure, based on nozzle manufacturer design/specs.		E	I	Consequences: 1.5.2.1
26. PT 302 / PSH 302 - monitoring for overpressurization and will trigger EMO and will shut down AOV 101-103 (H2 supply), AOV 201-212 (fueling valves), and AOV 301-312 (de-fueling valves).		E	I	Consequences: 1.6.1.1
27. PSV-301 set to the lesser of the hose MAWP or 2,600 psi. PSV will be sized for maximum flow through FCV-103 under worst case conditions (full open with maximum pressure)		E	I	Consequences: 1.6.1.1

Safeguards	Safety Critical Control	Category	Maximum Current Severity	Place(s) Used
28. Components downstream from FCV-103 are rated 35C above high temperature from the tanks of 85C.		E	I	Consequences: 1.8.1.1
29. If temperature < -40C are reached then system will stop dispensing should close within 5 seconds if normal operating temperatures are not achieved.		E	I	Consequences: 1.9.2.1
30. If TT 101-103 see a temperature < -40C then AOVs 101-103 will close after 5 seconds.		E	I	Consequences: 1.9.2.1
31. If system of any volume is filled with 100% air and then filled with H2 once the pressure exceeds 44 psig the UEL is exceeded and mixture is no longer flammable.		E	I	Consequences: 1.12.1.1
32. No ignition sources inside pipes		O	I	Consequences: 1.12.1.1
33. LOTO and zero-energy verification procedure must be developed for maintenance.		P	I	Consequences: 1.12.1.1, 3.12.1.1
34. No pressurized O2 source capable of creating flammable concentrations at pressure		O	I	Consequences: 1.12.1.1
35. HDHS is bonded and grounded reducing the likelihood of ignition		E	I	Consequences: 1.12.1.1, 1.15.2.1
36. N2 purge procedures as required by ASME code are		P	I	Consequences: 1.12.1.1

Safeguards	Safety Critical Control	Category	Maximum Current Severity	Place(s) Used
required prior to returning system to operation and are detailed in maintenance procedure.				
37. SOP includes checklist verification of all ancillary equipment connections		PPE	I	Consequences: 1.13.1.1
38. PPE including FR clothing, safety glasses,		PPE	I	Consequences: 1.13.1.1
39. All safety critical sensors will lead to fail safe condition upon failure or malfunction (e.g., out of range alarm, IO module watchdog), tripping EMO and shuts down.		E	I	Consequences: 1.14.1.1, 3.14.1.1
40. EMO, CGs, and flame detection relays are hardwired to remove control power to control valves, operating independent of PLC.		E	I	Consequences: 1.14.2.1, 3.12.1.2
41. All air operated shutoff valves return to safe state upon the loss of pressure from N2.		E	I	Consequences: 3.2.1.1, 3.14.1.1
42. PSL-702 low pressure N2 switch will trigger an alarm if low/no pressure.		E	III	Consequences: 3.2.1.1
43. All control system power is removed leading to valves close to safe state configuration.		E	I	Consequences: 1.15.2.1
44. Tank pressure gauges will be visible externally from the HDHS.		E	I	Consequences: 1.15.2.1

Safeguards	Safety Critical Control	Category	Maximum Current Severity	Place(s) Used
45. Select fans (based on H2 gas accumulation) will be C1D1 rated.		E	I	Consequences: 1.15.2.1
46. Control screen will remain off during loss of power		E		
47. Without power, less likely to have an ignition source in the HDHS.		O	I	Consequences: 1.15.2.1
48. C1D1 rated CG relay, UV/IR fire detection AND loss of ventilation relay will not allow non C1D1 equipment to power up until a safe atmosphere is verified.		E	I	Consequences: 1.15.2.1, 3.12.1.2
49. Control screen and beacon will remain off indicating loss of power to the system.		E	I	Consequences: 1.15.2.1
50. Process Beacon/light tree will be installed on the exterior of the HDHS to indicate process conditions (e.g., running, EPO/EMO)		E	I	Consequences: 1.15.2.1
51. C1D1 CG beacon/light tree will be installed on the exterior of the HDHS to indicate status of CG detection (combustible gas detection indication at/above high-high condition)		E	I	Consequences: 1.15.2.1, 3.12.1.2
52. Same as above, "EPO shutdown..." Safeguards.		O	I	Consequences: 1.15.2.2
53. CG and ventilation will be hardwired to autostart upon return of power.		E	I	Consequences: 1.15.2.2, 3.12.1.2

Safeguards	Safety Critical Control	Category	Maximum Current Severity	Place(s) Used
54. HDHS operators will be wearing FR clothing		PPE	II	Consequences: 1.16.1.1
55. Dispenser station internal cabinet is equipped with CG detection per code requirements		E	II	Consequences: 1.16.1.1
56. TT-401 A/B through TT-412 A/B, High-high temperature alarms triggering a process stop. (P-stop)		E	I	Consequences: 2.1.1.1, 2.3.1.1, 3.2.1.1, 3.3.1.1
57. TT-401 A/B through TT-412 A/B, High temperature alarm will notify trained operators.		E	I	Consequences: 2.1.1.1, 2.3.1.1, 3.2.1.1, 3.3.1.1
58. The failure mode of the tanks due to adiabatic overheating of the internal hydrogen is to damage the HDPE liner, causing hydrogen permeation (leak-through) of the carbon fiber structure. The tank will NOT rupture under adiabatic heating scenarios, confirmed with manufacturer, and through internally derived calculations and modeling. See zip file including simulations, notes from discussions with manufacturer, and literature papers.		E	I	Consequences: 2.1.1.1, 2.1.2.1
59. TT-401 A/B through TT-412 A/B, Low-low temperature alarms triggering a process stop. (P-stop)		E	III	Consequences: 2.3.1.1, 3.2.1.1, 3.3.1.1

Safeguards	Safety Critical Control	Category	Maximum Current Severity	Place(s) Used
60. TT-401 A/B through TT-412 A/B, Low temperature alarm will notify trained operators.		E	III	Consequences: 2.3.1.1, 3.2.1.1, 3.3.1.1
61. TPRD is a mechanical device that is only activated when TPRD exceeds temperature setpoint - manufacturers set point set by OEM		E	I	Consequences: 2.5.2.1, 2.8.1.1
62. All components of HDHS in this node are rated to >/=20000 psig. Highest calculated pressure of trapped with worst case ambient temperature is calculated <20,000 psig (-40C to 40C ambient)		E	I	Consequences: 1.6.2.1
63. FCV-103 flow control valve automatically controls based on feedback pressure decay rates of H2 Storage PT-401A through 412A that throttles venting		E	II	Consequences: 2.1.2.1
64. Allowable blowdown rate exceeded (calculated from pressure decay PT-401A through 412A and are tank specific), automatically shuts down AOV-401 thru 412		E	II	Consequences: 2.1.2.1
65. All TTs in the HDHS system have high-high temperature alarms and fault alarms triggering a process stop. (P-stop)		E	I	Consequences: 2.8.1.1
66. Trailer interior to be clad in non-flammable materials		E	I	Consequences: 2.8.1.1

Safeguards	Safety Critical Control	Category	Maximum Current Severity	Place(s) Used
(e.g., metal sheeting). Limited combustible material.				
67. Fueling station designed to NFPA 2 Gaseous Fueling Facilities and IFC		E	I	Consequences: 2.8.1.1
68. Separation distance of 30' from wildland areas per NFPA 1140 Standard for Wildland Fire Protection		E	I	Consequences: 2.8.1.1
69. Fueling protocol dictates flow rate based on inlet H2 temperature		E	I	Consequences: 2.8.2.1
70. Calculation of H2 gas density used to calculate tank specific SOC. SOC is designed to limit the end of fill density of H2 in tanks. Calculation is based on most conservative of PT-401A through PT-412A AND of TT-401A/B through TT-412A/B upon High-High SOC for the tank, P-Stop of the station		E	I	Consequences: 2.6.1.1
71. Station Dispenser H70 fueling MOP 87.5 MPa (12690 psig) and Station H70 fueling PSVs set at maximum of 96.25 MPa (13960 psig)		E	I	Consequences: 2.6.1.1
72. All non-tank components rated to a minimum of 20000 psig		E	I	Consequences: 2.6.1.1
73. H2 Storage Tanks are DOT rated and the minimum Test Pressure is rated to least the highest MAWP that can be set		E	I	Consequences: 2.6.1.1

Safeguards	Safety Critical Control	Category	Maximum Current Severity	Place(s) Used
for the Dispenser Station of 96.25 MPa (13960 psig)				
74. High-high and low-low alarms are always active for tanks, even if not intended for use		E		Consequences: 2.13.1.1
75. Out of Range fault activated when sensor unplugged		E		Consequences: 2.14.1.1
76. Password protecting control software to ensure safety system integrity according to cause and effects matrix		E		Consequences: 2.14.1.1
77. Ensure pathway to vent for fuel and de-fuel manifolds under power loss or loss of N2.		E	III	Consequences: 3.2.1.1
78. Trained and qualified workers performing maintenance operations.		T	II	Consequences: 3.3.1.1, 3.12.1.1
79. PSV-601 set to 2890 psig and venting to a safe location.		E	I	Consequences: 3.4.1.1, 3.12.1.1, 3.14.1.1
80. Rupture disc (BD-601) set at 3500 psi and venting to a safe location. Sized for chocked flow through 601 and 602 at MOP.		E	I	Consequences: 3.4.1.1, 3.14.1.1
81. PSH-601 high pressure (set higher than N2 cylinder) will EMO HDHS.		E	I	Consequences: 3.4.1.1, 3.12.1.1
82. Ensure sufficient air gap between process control and pneumatic operation in selection of H2 AOVs.		E	I	Consequences: 3.4.2.1, 3.6.1.1

Safeguards	Safety Critical Control	Category	Maximum Current Severity	Place(s) Used
83. N2 limited to volume of one cylinder (approximately 256 cu ft)		O	I	Consequences: 3.5.1.1, 3.6.1.1, 3.9.1.1
84. Trailer is not intended for human occupancy during operating conditions.		P	I	Consequences: 3.5.1.1, 3.6.1.1, 3.9.1.1
85. PSV-701 set at 125 psig, sized for the full flow of the regulator. Relief vented routed externally to safe location		E	I	Consequences: 3.6.1.1
86. Startup procedure will include N2 cylinder volume check		P	III	Consequences: 3.2.1.1
87. Lines 6001 to 6006 and components rated to -40C		E	I	Consequences: 3.9.1.1
88. Operating procedures require the doors to be left open when anyone is inside		P	I	Consequences: 3.9.1.1
89. Procedure requires PI-601 must read 0 psig before needle valves (NV-601 and NV-602) from N2 cylinder are opened		P	I	Consequences: 3.12.1.1, 3.12.1.2
90. If PSH-601 or PSV-601 are activated, operator must vent H2 system and conduct root cause analysis and determine cause of failure and mitigate before operation may continue.		P	I	Consequences: 3.12.1.1, 3.12.1.2
91. HDHS blown down less than 30 psig, remaining H2 in tanks up to 600 g of H2 (250 scf) in HDHS		P		Consequences: 4.1.1.1

Safeguards	Safety Critical Control	Category	Maximum Current Severity	Place(s) Used
92. Tanks must be mounted according to manufacturer's recommendation		E		Consequences: 4.1.1.1
93. Mounting of tubing and components to DOT codes and recommendations		E		Consequences: 4.1.1.1
94. Fan louvers provide passive ventilation		O		Consequences: 4.1.1.1

HAZOP Recommendations Summary

Recommendations	Place(s) Used	Maximum Current Risk Ranking	Maximum Residual Risk Ranking	Responsibility	Resolution	Resolution Date
1. Evaluate if receptacle life is a concern and add receptacle cycle life counter within control system via PT-101, PT-102 and PT-103	Consequences: 1.3.2.2					
2. Recommend operators PPE to include personal CG/O2 detector and FR clothing and FR gloves when dispensing or conducting maintenance of HDHS.	Consequences: 1.3.2.2, 1.13.1.1, 3.5.1.1, 3.6.1.1, 3.9.1.1	Moderate	Low			
3. Implement alarm on PT-101 or PT-103 high pressure alarm	Consequences: 1.3.2.2					
4. Interlock all tank vent valves with NV-104 being open	Consequences: 1.3.3.1, 1.3.3.2	Low	Low			
5. Conduct leak check of H2 piping, with inert gas, up to isolation valve (NV-401 through NV-412) after every relocation and prior to testing of the HDHS.	Consequences: 1.5.1.1, 2.5.1.1	Moderate	Low			
6. Utilize torque seal and/or indicator to identify if hardware has loosened during transportation.	Consequences: 1.5.1.1, 2.5.1.1	Moderate	Low			
7. After fueling event, decouple the nozzle and then vent down the HDHS manifold prior to recirculating. Close AOV 101-103, limited pressure	Consequences: 1.5.2.1					

Recommendations	Place(s) Used	Maximum Current Risk Ranking	Maximum Residual Risk Ranking	Responsibility	Resolution	Resolution Date
8. After fueling event, procedure for operator to initiate (end of fill sequence), reducing H2 pressure and volume by limiting amount of H2 available by blocking up to receptacles. REVISIT TO DETERMINE IF PSV SHOULD BE INCLUDED AND REALITY OF THIS CONTROL.	Consequences: 1.5.2.1					
9. Relocate NV-104 downstream from de-fueling vent prior to FCV-103 to use an administrative lock that will be placed during shutdown.	Consequences: 1.13.1.1	Moderate	Low			
10. Include an electrical interlock which will close a loop only once the vent mast is installed (manual connection).	Consequences: 1.13.1.1	Moderate	Low			
11. Conduct leak check of H2 piping, with inert gas, up to isolation valve (NV-401 through NV-412) after every relocation and prior to testing of the HDHS. This will be the first indication if the vent mast is not installed properly, leaking only inert gas.	Consequences: 1.13.1.1	Moderate	Low			
12. Use a safety rated PLC control OR use a redundant watchdog system (if one system goes down or loses communication the whole process shuts down).	Consequences: 1.14.2.1	Low	Low			
13. SOP will include do not enter HDHS upon power failure and loss of monitoring equipment.	Consequences: 1.15.2.1	Low	Low			
14. Upon reenergization of HDHS, if C1D1 rated CG relays identified unsafe atmosphere exterior indicator signal will be activated to alert workers to hazardous atmosphere.	Consequences: 1.15.2.2	Low	Low			
15. SOP no worker should open and/or enter HDHS until safe atmosphere is safely confirmed.	Consequences: 1.15.2.2	Low	Low			
16. Ensure ventilation equipment is C1D1 rated	Consequences: 1.15.2.2	Low	Low			
17. SOP no worker should open and/or enter HDHS until safe atmosphere is safely confirmed.	Consequences: 1.15.2.1	Low	Low			
18. Workers should wear personal CG monitors	Consequences: 1.16.1.1	Low	Low			
19. TPRD vent stack draining only occurs when HDHS is below 30 psig. Maintenance procedures should not allow draining vent mast at tank pressures above the travel pressure of 30 psig and verifying the draining cap has been installed correctly.	Consequences: 2.5.2.1	Moderate	Low			
20. Consider adding if TT 101 through 103 see a temperature < -45C the triggers immediate shutdown AOVs 101 thru 103	Consequences: 1.6.2.1, 1.9.2.1	Low	Low			

Recommendations	Place(s) Used	Maximum Current Risk Ranking	Maximum Residual Risk Ranking	Responsibility	Resolution	Resolution Date
21. Consider early shutdown of fueling valve AOV-101, 102, 103 based on H2 high temperature from station as prescribed in developing Fueling Protocol	Consequences: 2.8.2.1	Low				
22. Consider adding P-stop to PSL-702	Consequences: 3.2.1.1	Routine				
23. Determine if it is necessary to vent the fuel and de-fuel manifolds under certain safety conditions. Will determine how to proceed with existing safeguards 84.	Consequences: 3.2.1.1	Routine				
24. Industry standard H2 process control valves include designs similar to autoclave engineer SM model valves.	Consequences: 3.4.2.1	Low				
25. Develop response procedure for contaminated N2 cylinder system.	Consequences: 3.12.1.1, 3.12.1.2	Low	Low			

Appendix F. CHSS Depressurization Rate Derivations

The SAE J2601-5 fueling protocol regulates the fueling speed to prevent medium- and heavy-duty vehicle storage systems from overheating and overfilling. However, there is no guidance on how quickly hydrogen can be discharged from those storage systems. For this reason, NREL evaluated the maximum hydrogen depressurization rates for the HD HyStEP device to protect the individual inner tank materials from being subjected to temperatures below -40°C. Below are the steps involved in the evaluation and modeling process to determine the rates for various CHSS configurations.

- Configure the HD HyStEP tank systems in NREL's Hydrogen Filling Simulation (H2FillS) model.

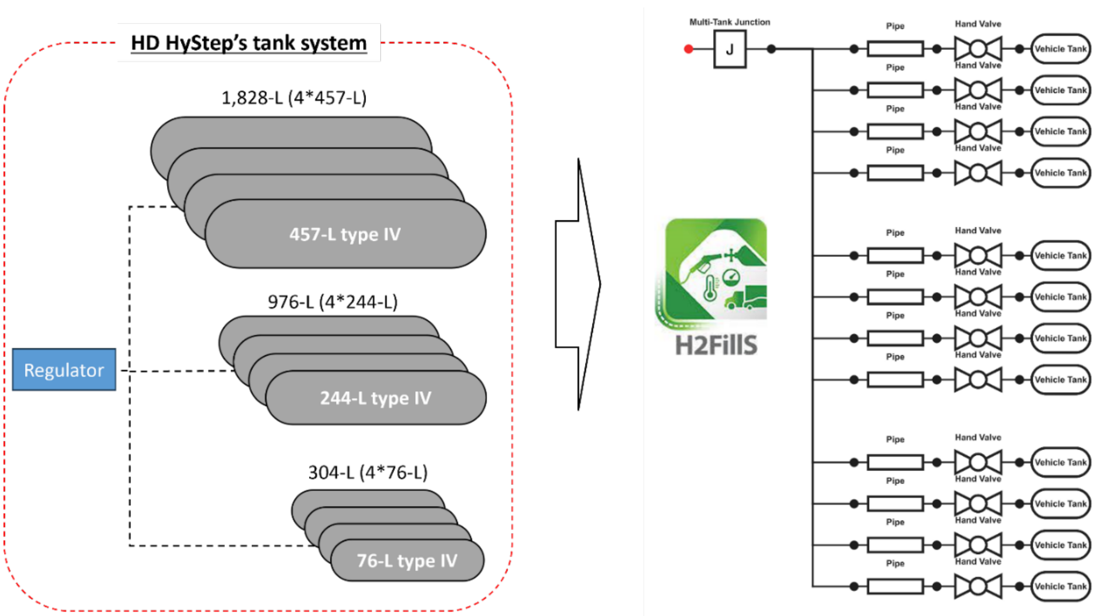

Figure 1. HD HyStEP configuration implemented into H2FillS

- Set the initial hydrogen pressure in the entire system, including the tanks, to 87.5 MPa. This is the maximum allowable working pressure set by the SAE J2601-5 fueling protocol.
- Set the initial temperature in the entire system, including the tanks, to -20°C to 50°C in 10°C increments. The material (e.g., piping and tank walls) temperatures are assumed to be equivalent to the hydrogen at the beginning of the depressurization process.
- Set a depressurization rate (MPa/min) to the regulator outlet and simulate the temperature drops in the individual CHSS tanks.
- The model records the tank temperature when the regulator outlet pressure reaches 0.5 MPa. If the tank temperature was not within the -35°C to -40°C bounds (highlighted yellow in Figure 2), the simulation was repeated with a new depressurization rate until the model found a rate that placed all final tank temperatures within -35°C to -40°C.

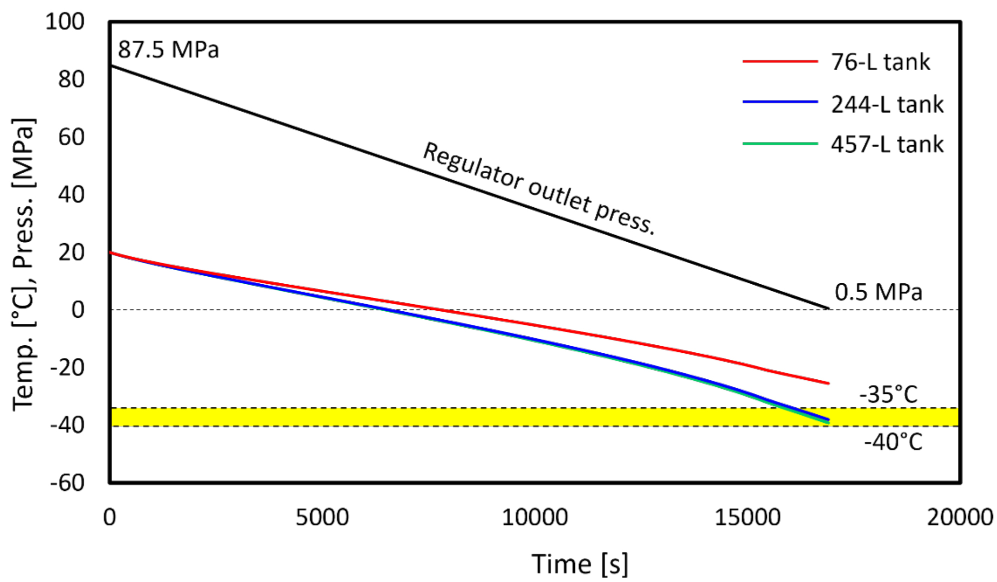

Figure 2. Depressurization simulation results with a three-tank system consisting of single 76 L, 244 L, and 457 L tanks

- Figure 2 shows a simulation result obtained with an HD HyStEP device configuration consisting of single 76 L, 244 L, and 457 L tanks. It can be seen that the hydrogen bulk gas temperatures in the 244 L and 457 L tanks decreased more rapidly than the bulk gas temperatures within the 244 L and 76 L tanks. Therefore, the hydrogen temperature in the 457 L (largest) tank tends to cool quicker than the 244 L and 76 L tank and its depressurization rate is the limiting factor for this example configuration. If a 457 L tank is included, the depressurization rate needs to be slower to ensure the bulk gas temperature in any of the tanks never reaches the low limit of -40°C.